

Multi-Agent Combinatorial Contracts*

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Abstract

Combinatorial contracts are emerging as a key paradigm in algorithmic contract design, paralleling the role of combinatorial auctions in algorithmic mechanism design. In this paper we study natural combinatorial contract settings involving teams of agents, each capable of performing multiple actions. This scenario extends two fundamental special cases: the single-agent combinatorial action model of [18], and the multi-agent binary-action model of [4, 19].

This setting presents fundamentally different challenges compared to the previous special cases, as it lacks key properties that have been crucial for resolving these scenarios. To navigate these challenges, we develop a broad set of novel tools that allow us to establish approximation guarantees for this setting.

Our main result is a constant-factor approximation for multi-agent multi-action problems with submodular rewards, given access to value and demand oracles. This result is tight: we show that this problem admits no PTAS (even under binary actions). As a byproduct of our main result, we devise an FPTAS, given value and demand oracles, for single-agent combinatorial action scenarios with general reward functions, which is of independent interest. Finally, we show that for subadditive rewards, perhaps surprisingly, the gap between the optimal welfare and the principal's utility scales logarithmically (rather than linearly) with the size of the action space.

1 Introduction

A fundamental problem in economics concerns situations, where a principal, delegates the execution of a project to a team of agents (see, e.g., [36]). If the project succeeds the principal enjoys some reward. Naturally, the team will consist of several agents, and each agent will take several actions towards a successful completion of the project. Different combinations of agents and their chosen actions, may affect the success probability in different ways. In addition, it's natural to assume that agents incur some cost for the actions they take, and that they are only willing to exert effort (take costly actions), if they get paid for their work. If the principal can observe the individual agents' choices of action, he could easily define a compensation scheme to have agents choose the desired actions. However, in many practical situations it is impossible or impractical for the principal to monitor the agents' actions. In this case, any payments the principal makes to the agents will be a function of the project's outcome alone. This, in turn, creates some intricate strategic interactions between the agents. Specifically, agents' best responses will now depend on the choices of the other agents, and agents may free-ride on the effort exerted by others.

Model. A possible abstraction of this problem is as follows. A principal interacts with a set of n agents. Each agent can take any subset S_i of actions A_i . We write $A = \bigcup A_i$ for the set of actions, and $S = \bigcup S_i$ for a choice of actions by the agents. Note that we may have $S_i = \emptyset$, in which case agent i does not take any action. Each action $j \in A$ comes with a cost c_j for the respective agent. Costs are additive, so that $c(S) = \sum_{j \in S} c_j$. In particular, $c(\emptyset) = 0$, so that not taking any actions comes with no cost. There are two possible outcomes: success or failure of the project. The principal's reward in case of success is r , and it is zero otherwise. Each set of actions S induces a success probability $f(S)$ of the entire project. The success probability f is normalized, so that $f(\emptyset) = 0$, and

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monotonically non-decreasing, so that $f(S) \leq f(T)$ for any $S \subseteq T$. A particular focus of this work is on the case, where f is submodular (see Section 2). The principal cannot observe the agents' actions, only the final outcome. The principal incentivizes the agents through a (linear) contract $\vec{\alpha} = (\alpha_1, \dots, \alpha_n)$, which specifies the fraction of r to be paid to each agent i in case of success. Consider a choice of actions S . Then agent i 's expected utility for the set of actions $S_i = S \cap A_i$ under contract $\vec{\alpha}$ is given by $\alpha_i f(S) \cdot r - c(S_i)$. Note that agent i 's utility depends on the choice of actions $S_{-i} = S \setminus S_i$ of the agents other than i . We say that a set of actions S is a (pure) Nash equilibrium to contract $\vec{\alpha}$, if for each agent i , $S_i \in \arg \max_{T \subseteq A_i} (\alpha_i f(T, S_{-i}) \cdot r - c(T))$. The principal's goal is to maximize her expected profit, given by $(1 - \sum_i \alpha_i) f(S) \cdot r$, when agents choose a set of actions S that is a best response to the announced contract.

Challenges. This multi-agent multi-action contracting model is the joint generalization of two fundamental combinatorial contracting models. The first is the single-agent combinatorial action model of [18], which is the special case with $n = 1$ agent. The second is the multi-agent binary-action model of [4, 19], which corresponds to the case where $A_i = \{i\}$ for each agent $i \in [n]$.

In both special cases the principal seeks to find the best set (of actions, or agents) out of exponentially many possible sets. However, the nature of the problems and hence the challenges are quite different. For instance, while in the multi-agent binary-action model all sets of agents can be incentivized, and it's clear what the best way to incentivize each set of agents is; not all sets of actions can be incentivized in the single-agent combinatorial actions model, and identifying which sets can be incentivized is one of the main difficulties. In both cases, some form of monotonicity comes to the rescue. In the single-agent combinatorial actions model it is that the success probability under the agent's best response is increasing in the scalar α . In the multi-agent model with binary actions (and submodular rewards), it is that when we drop an agent (while keeping the contracts fixed) the other agents who exerted effort before, would still exert effort.

The combined model studied in this paper, inherits the challenges of both special cases, but as it turns out, the challenges in the combined problem extend greatly beyond those encountered in the special cases. Specifically, given a set of agents and their actions, it's difficult to decide whether this profile can be incentivized at all; and even if it is, it's not clear what's the best way to achieve this.

In particular, the combined model loses the monotonicity properties that were instrumental in solving the special cases. For example, over- or under-paying an agent, can have some counter intuitive consequences on the resulting equilibrium. The following demonstrates this, by showing that in the combined model, unlike in the multi-agent binary-action model (with submodular f), reducing the payment to one agent may cause another agent to exert less effort.

EXAMPLE 1.1. (LACK OF MONOTONICITY) Consider a principal-agent setting with one principal and two agents. The principal's reward is $r = 1$. The action sets are $A_1 = \{1, 2\}$, $A_2 = \{3\}$. The costs are $c_1 = 0.1$, $c_2 = 0.24$, $c_3 = \epsilon$ for $\epsilon > 0$ small enough. The success probabilities are $f(\{1\}) = \frac{7}{16}$, $f(\{2\}) = \frac{3}{4}$, $f(\{3\}) = \frac{1}{4}$, $f(\{1, 3\}) = \frac{9}{16}$, $f(\{1, 2\}) = \frac{7}{8}$, $f(\{2, 3\}) = f(\{1, 2, 3\}) = 1$. Note that the success probability function defined this way is submodular but not gross substitutes (see Section 2).

- The principal utility is maximized by incentivizing $\{2, 3\}$ with contract $\vec{\alpha} = (\frac{8}{25}, 4\epsilon)$, which gives the principal a utility of $1 - \frac{8}{25} - 4\epsilon = \frac{17}{25} - 4\epsilon = 0.68 - 4\epsilon$.
- The optimal way to incentivize $\{1\}$ is by using contract $\vec{\alpha} = (\frac{8}{35}, 0)$, which gives the principal a utility of $(1 - \frac{8}{35}) \frac{7}{16} = \frac{27}{80} = 0.3375$
- The optimal way to incentivize $\{2\}$ is by using contract $\vec{\alpha} = (\frac{56}{125}, 0)$, which gives the principal a utility of $(1 - \frac{56}{125}) \frac{3}{4} = \frac{207}{500} = 0.414$
- The optimal way to incentivize $\{1, 2\}$ is by using contract $\vec{\alpha} = (\frac{4}{5}, 0)$, which gives the principal a utility of $(1 - \frac{4}{5}) \frac{7}{8} = \frac{7}{40} = 0.175$.
- The optimal way to incentivize $\{3\}$ is by using contract $\vec{\alpha} = (0, 4\epsilon)$, which gives the principal a utility of $\frac{1}{4}(1 - 4\epsilon) = \frac{1}{4} - \epsilon = 0.25 - \epsilon$.

We note that it is also possible to incentivize $\emptyset$ and $\{1, 3\}$ but these give smaller principal utility. Observe that when agent 2 is dropped, then we need to pay more to agent 1 to incentivize him to take action 2 ($\frac{56}{125} > \frac{8}{25}$). This

in particular means that when we keep $\alpha_1 = \frac{8}{25}$ and set $\alpha_2 = 0$, then agent 2 will take no action, while agent 1 will do $\{1\}$. So paying one agent (agent 2) less causes another agent (agent 1) to do less.

Another challenge in the combined model is that there can be multiple equilibria; and there is a gap between the best and the worst equilibrium. Multiplicity of equilibria is not an issue in the special cases of our model that have been studied before. In the model of [18] the agent's choice is unique up to tie-breaking at the points where the agent's best response to the contract changes. In the model of [19], the equilibrium is unique (for submodular f , under mild assumptions). Moreover, the fact that there is a clear choice of contracts to incentivize a set of agents to exert effort, also made prior work focus on not only computing a contract for each agent but also a recommended action.

EXAMPLE 1.2. (MULTIPLICITY OF EQUILIBRIA) Consider a setting with two agents where $A_1 = \{1, 2\}$, and $A_2 = \{3, 4\}$. The actions are symmetric within each action set of the agents (i.e., the success probability depends only on the number of actions taken by each agent). Let the success probability $f(x, y)$ be as defined in Table 1, where x and y denote the number of actions taken by agent 1 and 2, respectively.

$x \setminus y$	0	1	2
0	0	0.3	0.6
1	0.4	0.65	0.85
2	0.7	0.9	0.9

Table 1: The success probability function $f(x, y)$, where x is the number of actions taken by agent 1, and y is the number of actions taken by agent 2.

It is easy to verify that f is submodular but not gross-substitutes.

The cost of all actions is 0.005. The optimal contract is then to incentivize agent 1 to take two actions and agent 2 to take one action by offering the contract of $\alpha = (0.025, 0.02)$ leading to a success probability of 0.9, and a principal utility of 0.8595. This is the optimal contract since this is the minimal way to incentivize an action profile with a success probability of more than 0.85. However, under this contract, there is another equilibrium where agent 1 takes one action and agent 2 takes two actions, leading to a utility of 0.81175 for the principal. Moreover, no contract can guarantee better than 0.81175 utility under all equilibria as any contract that incentivizes the action profile where agent 1 takes two actions and agent 2 takes one action, also incentivizes the profile where agent 1 takes one action and agent 2 takes two actions (and the profile where both agents take two actions cannot be incentivized).

This example also establishes limits to guarantees that can be established if one seeks worst-case guarantees that apply under any equilibrium. Namely, it shows that there is a constant $\gamma < 1$ for which we cannot guarantee more than a γ fraction of the utility under the best equilibrium of the best contract, even when ignoring computational aspects.

1.1 Our Results We start with a simple but important observation (which was independently observed in [16] for the multi-agent binary-action model). Namely, for any fixed contract $\vec{\alpha}$, the induced game among the agents is a potential game (see Section 2, Proposition 2.1). This shows that any contract $\vec{\alpha}$ admits at least one (pure) Nash equilibrium. Moreover, from the particular form of the potential function, it's also clear that one can find *some* equilibrium S with a single demand query to f .

Constant-factor approximation. Our first main result is a polynomial-time algorithm that gives constant approximation to the optimal principal's utility for any setting with submodular success probability function f , under value and demand oracles. Notably, our algorithm finds a contract $\vec{\alpha}$, so that *any* equilibrium S of this contracts yields a high utility for the principal. This worst-case type guarantee is significantly stronger than simply ensuring the existence of a good equilibrium.

MAIN THEOREM 1.1. (SEE SECTIONS 3, 4, 5) *There exists a poly-time algorithm, issuing poly-many value and demand queries that, for every multi-agent multi-action contract setting with submodular reward function, computes a contract $\vec{\alpha}$ such that every equilibrium of $\vec{\alpha}$ gives a constant approximation to the optimal principal's utility.*

Our proof of this theorem requires several new insights, and new techniques. The proof reduces the problem to one of two cases: Either no agent is large, or only a single agent is incentivized. (We remark that although a similar reduction appeared in [19], it's all but obvious that this would also hold beyond binary actions, because of the non-monotonicity illustrated in Example 1.1).

For the “no agent is large” case we give a constant-factor approximation with access to a value oracle. Our algorithm relies on a novel concept of subset stability, and a new Doubling Lemma, which links this concept to contracts and equilibria. Our approach for this case also uncovers an interesting connection to a particular form of bundle prices. We use the particular form of these bundle prices, to argue that we can efficiently optimize over these prices.

To address the “only a single agent is incentivized” case, we first show how to get an FPTAS for the single-agent problem with access to a value and demand oracle, even for general reward functions. The key difficulty in showing this result is to argue that the optimal contract is strictly bounded away from 1. Our argument for this exploits the (known) upper bound on the utility-to-welfare gap [25]. We then show how to extend and robustify the resulting contracts for the setting with multiple agents, using subset stability and the Doubling Lemma, losing only an additional constant factor.

The FPTAS that we give for the single-agent problem is of independent interest. It improves upon a weakly FPTAS that has been established by [18]. It is also tight in that it is known that, even for submodular f computing an optimal contract is NP-hard [18], and may require exponentially many demand queries [20].

We remark that Main Theorem 1.1 also implies the first constant-factor approximation for the multi-agent multi-action problem with gross substitutes f , with value oracle access only. This problem is known to be NP-hard, even for additive f and binary actions [19].

Tightness/impossibility. Our second main result shows that Main Theorem 1.1 is essentially tight. Namely, it shows that a constant approximation that is established for submodular f with value and demand oracle access is the best we can hope for, even with binary actions and even when the principal can choose the equilibrium.

MAIN THEOREM 1.2. (SECTION 6) *No PTAS exists for the optimal contract problem in multi-agent contract settings with submodular reward and with value and demand oracles, even in the special case of binary actions, and even under the best equilibrium.*

This theorem strengthens two previous impossibility results for the multi-agent binary-action problem. Specifically, for submodular rewards f , it was known that there cannot be a PTAS, assuming value oracle access only [27]. Under demand oracle access, it was only known that there could be no PTAS for the broader class of XOS success probability functions f [19].

Our proof of the impossibility theorem differs quite significantly from these previous impossibilities: Rather than hiding a valuable set, it hides a set where the slopes (= marginals) are higher (and hence the payments are smaller).

Second best vs. first best. The social welfare of an action set S is the difference between its reward and cost, namely $f(S) - c(S)$. Our final set of results concerns the gap between the optimal principal's utility from a contract and the optimal social welfare of an instance. This gap is also known as second-best vs. first-best in economics.

MAIN THEOREM 1.3. (SEE SECTION 7) *The following guarantees hold with respect to the first-best:*

- *For a single agent with subadditive reward function f , the gap between the optimal social welfare and the best principal's utility is upper bounded by the number of actions, m . Moreover, the same holds for the best equilibrium in multi-agent settings with subadditive reward f .*
- *Combined with Main Theorem 1.1, for submodular rewards f , the best contract obtains an $\Omega(\frac{1}{m})$ fraction of the optimal social welfare under every equilibrium.*

Two remarks are in order. First, this result may be surprising in light of previous results. Specifically, [25] show that the gap between the optimal social welfare and the best principal utility is bounded by the number of critical α 's, while [18] showed that even for submodular reward functions the number of best responses can be exponential in m . Thus, our result shows that the actual bound is exponentially better than what one could expect given [25, 18]. Second, this result, beyond being interesting in its own right, implies an improved query complexity of our FPTAS for the single-agent combinatorial actions setting when rewards are subadditive.

1.2 Our Techniques

Constant approximation and dealing with multiplicity of equilibria. We first present and discuss the notion of subset stability (Definition 3.1) along with the Doubling Lemma (Lemma 3.1), which are crucial components of our approximation algorithm. Recall that, in an equilibrium of a contract, no agent can increase their utility by deviating to any other choice of actions. We introduce a weaker condition called subset stability: No agent wants to deviate to any subset of the actions. This condition turns out to be extremely powerful because of our Doubling Lemma (Lemma 3.1). This lemma asserts that if S is subset stable with respect to a contract $\vec{\alpha}$, then any equilibrium $S^\dagger$ of a contract $2\vec{\alpha} + \vec{\epsilon}$ fulfills $f(S^\dagger) \geq \frac{1}{2}f(S)$. Therefore, in a way, it is sufficient to compute a contract admitting a subset stable set of high value. The advantage of subset stability (compared to standard stability required in equilibria) is that it admits useful properties, such as closure under removal of actions, making it easier to ensure.

Our method for designing an approximation algorithm involves taking the better of two contracts. The first contract we compute approximates the best contract in the case that no agent is “large”, meaning that, for the optimal contract $\vec{\alpha}^*$ and the optimal equilibrium S^* of contract $\vec{\alpha}^*$, for each agent $i \in [n]$, $f(S_i^*)$ is bounded away from $f(S^*)$. In this case, we design an algorithm to compute a contract $\vec{\alpha}$ and a subset stable set S such that $\sum_{i \in [n]} \alpha_i \leq \frac{1}{4}$ and $f(S) = \Omega(1) \cdot f(S^*)$. Then our Doubling Lemma shows that $2\vec{\alpha} + \vec{\epsilon}$ is a good approximation of the optimal contract. Furthermore, while subset stability is easier to ensure than standard stability, it still requires verifying exponentially many conditions. To mitigate this problem, we identify a sufficient condition for subset stability. Morally, it suffices to find a set that, for some $\gamma > 0$, maximizes

$$f(S) - \gamma \sum_{i \in [n]} \sqrt{\sum_{j \in S_i} c_i}.$$

This can be interpreted as finding a demand set with respect to a particular type of (non-additive) bundle prices. For this latter problem, we devise an approximation algorithm.

The second contract we compute approximates the best contract in which only one agent is incentivized, meaning that, in the optimal contract $\vec{\alpha}^*$, $\alpha_i^* = 0$ for all but one agent i . We first assume that there is only a single agent, which is the setting studied in [18]. We show that this problem admits an FPTAS even for general f ; so far only algorithms running in weakly polynomial time were known. To this end we show that, in the optimal contract for this agent, α is bounded away from 0 and 1. The bounds we derive exploit (known) bounds on the gap between first best and second best, in particular the fact that the principal can achieve a utility of at least $\frac{1}{2^m} \max_{S'} (f(S') - c(S'))$. We then discretize the range of possible α 's, and determine the best contract in this polynomial-sized set. This leads to a query complexity of $O(m^2)$. (In Proposition 7.1, we sharpen the lower bound on the utility the principal can achieve with a linear contract when f is subadditive to $\frac{1}{m} \max_{S'} (f(S') - c(S'))$. This yields an improved query complexity of $O(m \log m)$.) Transitioning back to the multi-agent setting, other equilibria may emerge, involving additional agents (beyond the intended one) taking actions. By appealing once more to the concept of subset stability and the Doubling Lemma, we show that the loss to the principal's utility due to these agents can be bounded.

The final step is to argue that the optimal principal utility is well approximated (to within a constant factor) by the two cases: no agent is large, and only a single agent is incentivized. This last step involves demonstrating that in the presence of a large agent in the optimal solution, incentivizing a single agent is sufficient. Due to the lack of monotonicity observed in Example 1.1, this property is not as straightforward as it may appear. Instead, we use the facts that every equilibrium is also subset stable and that subset stability is preserved under removal of actions, and then apply our Doubling Lemma.

Impossibility by hiding a set with high marginals. To establish Main Theorem 1.2, our impossibility for the multi-agent setting with binary actions, we construct a family of hard instances based on a carefully chosen function on two types of agents, “good” agents which we want to incentivize in the optimal contract, and “bad” agents which we don’t want to exert effort. While this general approach was also adopted by former constructions [19, 27], our approach differs from these prior approaches in a fundamental way. Namely, rather than hiding a set of high value, we hide a set with high marginals.

In our construction, the value of a set that contains a higher fraction of good agents is no higher than the value of a set of agents of the same size with a lower fraction of good agents. Moreover, the value is strictly lower, unless the fraction of the good agents is small enough, or the size of the set of agents is large enough. This construction makes the marginal contribution of every good agent to the set of good agents higher, which in turn decreases the payments necessary to incentivize them. The good set of agents (and all its subsets) remains hidden, as demand oracles are good at finding sets with higher values as opposed to sets with lower values.

Bounding the welfare gap action by action. We first analyze the case of a single agent. For this case, instead of considering all critical α ’s, we split the interval of contracts $[0, 1]$ to sub-intervals in places where new actions enter the best response of the agent for the first time. By the fact that the welfare can be written as the integral over the success probability as a function of the contract α , and with the subadditivity of the success probability function, this split allows us to bound the social welfare as a sum of expressions, one for each action. On the other hand, the minimal contract that incentivizes an action j , is one of these points in the split, and achieves the corresponding expression in the bound of the welfare. This gives us the bound on the gap to the welfare of at most the number of actions.

To extend it beyond the single agent setting, we design a randomized contract that incentivizes only a single agent. We choose the agent to incentivize with a probability that is proportional to the number of actions he has. This random choice balances the contributions of the agents since the approximation for the single agent is proportional to the inverse of the number of actions, which gives us the approximation of the number of actions. The optimal contract achieves at least the utility of the randomized contract we analyze.

1.3 Related Work

(Algorithmic) contract theory. Contract theory is one of the pillars of microeconomic theory [44, 35, 31], and has been recognized by the 2016 Nobel Prize to Hart and Holmström. Motivated by the fact that more and more of the classic applications of contract theory are moving online and growing in scale, recent work has started to explore contracts from an algorithmic perspective (see, e.g., [4, 34, 25]).

A central topic in the emerging field of algorithmic contract theory are “combinatorial contracts.” The multi-agent binary-action model is pioneered and explored in [4, 19], with [19] exploring the hierarchy of complement free success probability functions. The single-agent combinatorial action model was proposed and first studied in [18]. Additional work that provides both positive and negative results for these two models includes [27, 16, 20, 22]. As a strict generalization of both these models, our work inherits the hardness results that have been established for the two special cases.

Most closely related to our work is a recent working paper by [8]. This paper studies a related but different multi-agent contracting setting with multiple actions per agent. This work, unlike our work, assumes that each agent i has a fixed set of actions A_i , and can take any action $j \in A_i$. Moreover, any action profile (a choice of a single action for each agent) induces a probability distribution over possible outcomes (not necessarily binary). A main contribution of this work is the introduction of the concept of randomized contracts for this setting (also see work on typed contracts below). They show that randomized contracts can be arbitrarily better than deterministic ones, and give an algorithm that finds a $(1 + \epsilon)$ -approximate randomized contract in time polynomial in $O(1/\epsilon)$ and the description size of the problem. The work also shows a reduction from the multi-agent setting to a single-agent setting (via virtual costs), and uses this to show an approximation guarantee for linear contracts.

Two additional directions in combinatorial contracts include [26] and [12]. The former studies a single agent contracting setting with combinatorial outcome space [26], the latter studies a related (but different) multi-agent contracting problem, where each agent’s action leads to an observable individual outcome which an agent’s payment can depend on [12].

The studies of [25, 5] explore the gap between linear contracts and optimal contracts, as well as between the best contract and the optimal social welfare. There is a significant amount of work on typed contracts, including

[32, 10, 11, 2, 1]. There is also work that considers the problem of learning contracts, especially from an online learning, no-regret perspective [34, 23, 46].

Combinatorial optimization. On a technical level, our work is related to work on combinatorial optimization, in particular work on combinatorial auctions. Here, gross substitutes appear as a natural frontier for economic but also algorithmic reasons (e.g., [42]). In a seminal paper, [28] provides a constant-factor approximation for the welfare maximization problem in combinatorial auctions with submodular, XOS, and subadditive bidders. An exciting line of work, seeks truthful approximation algorithms for submodular and XOS bidders [17, 3]. A different line of work has approached this problem, by analysing the Price of Anarchy of simple combinatorial auctions, such as combinatorial auctions with item bidding [14, 29]. Related set-valued optimization problems come up in prophet inequalities and online posted price mechanisms [30, 21, 24, 15]. There are also polynomial-time constant-factor approximation results for truthful revenue maximization with unit-demand bidders [13], additive bidders [45], and XOS bidders [9]. An important difference between these works on combinatorial allocation problems and our work, in addition to the different incentive problems arising due to the hidden action aspect of the problem, is that in our case the set-valued function f is not additively separable across agents.

Optimizing the effort of others. More broadly, our work contributes to an emerging frontier in computer science, seeking to optimize the effort of others. Additional domains within this frontier encompass strategic classification (e.g., [38, 6]), optimal scoring rule design (e.g., [33, 40]), and delegation (e.g., [37, 7]).

2 Model

Principal-agent model. We consider the case where a single principal (she, her) delegates the execution of a project to n agents (he, his). We focus on the binary-outcome case, where the project can either succeed or fail. We refer to these two outcomes as 0 and 1, respectively, and let $\Omega = \{0, 1\}$. The principal receives a reward of $r > 0$ for outcome 1, and a reward of zero for outcome 0. A simple scaling argument shows that it is without loss of generality to assume that $r = 1$, and this would be our default assumption. However, for some examples and constructions it will be convenient to work with general $r > 0$. Each agent $i \in [n]$ has a set of actions A_i that he can choose from. We assume that the sets of actions are disjoint, i.e., $A_i \cap A_{i'} = \emptyset$ for any pair of agents $i \neq i'$ with $i, i' \in [n]$. We write $A = \bigcup_{i \in [n]} A_i$ for the set of actions. We use $m = |A|$ to refer to the total number of actions.

Agent i can take any subset $S_i \subseteq A_i$ of actions (including $S_i = \emptyset$). We write $S = \bigcup_{i \in [n]} S_i \subseteq A$ for the set of actions taken by the agents. Each action $j \in A$ comes with a (non-negative, real) cost $c_j \in \mathbb{R}_+$. The cost of a set of actions $S \subseteq A$ is additive, i.e., $c(S) = \sum_{j \in S} c_j$ (with $c(\emptyset) = 0$). A success probability function $f : 2^A \rightarrow [0, 1]$ maps each set of actions to a success probability. When the reward for success is normalized to $r = 1$, the success probability coincides with the expected reward of a set of actions. We thus also refer to f as the reward function.

Contracts and utilities. In general, a contract defines outcome-contingent transfers from the principal to the agent. In the binary-outcome case that we focus on, it is without loss of generality to pay the agents only on success (see Observation II.1 in [18]). It is therefore without loss of generality to consider linear contracts. A linear contract $\vec{\alpha} = (\alpha_1, \dots, \alpha_n)$ specifies, for each agent $i \in [n]$, a fraction $\alpha_i \in [0, 1]$ of the reward that should be transferred to the agent upon success. Suppose given contract $\vec{\alpha}$ the agents choose actions S , then the principal's utility is given by

$$u_p(S, \vec{\alpha}) = \left(1 - \sum_{i \in [n]} \alpha_i\right) \cdot f(S) \cdot r.$$

In turn, agent i 's utility is given by

$$u_i(S, \vec{\alpha}) = \alpha_i \cdot f(S) \cdot r - \sum_{j \in S \cap A_i} c_j.$$

Note that agent i 's utility depends on his own actions $S_i = S \cap A_i$, but also on the actions $S_{-i} = S \cap (\bigcup_{i' \neq i} A_{i'})$ chosen by the other agents. To highlight which actions are controlled by which agents, we sometimes write $f(S_i, S_{-i})$ to mean $f(S_i \cup S_{-i})$, and similarly for other functions. A choice of actions S is a (pure) Nash equilibrium

for contract $\vec{\alpha}$, if for each agent $i \in [n]$ and all $S'_i \subseteq A_i$ it holds that

$$u_i(S_i, S_{-i}, \vec{\alpha}) \geq u_i(S'_i, S_{-i}, \vec{\alpha}).$$

We note that the special structure of the games we consider ensures the existence of a pure Nash equilibrium for any $\vec{\alpha}$. This can be shown by observing that $\phi(S) = f(S) - \sum_{i \in [n]} c(S_i)/\alpha_i$ is a potential function. (If $\alpha_i = 0$ we define $c_j/\alpha_i = \infty$ when $c_j > 0$, and we let $c_j/\alpha_i = 0$ when $c_j = 0$.) For the special case of binary actions, this observation has been made by [16]. A proof appears in Appendix A.

PROPOSITION 2.1. (SEE ALSO [16]) *Every contract $\vec{\alpha}$ admits at least one pure Nash equilibrium S .*

The optimization problem. Our goal is to compute a contract $\vec{\alpha}$ and a choice of actions S (to be recommended to the agents) such that the choice of actions S constitutes a Nash equilibrium for contract α and such that

$$u_p(S, \vec{\alpha}) \geq \kappa \cdot \sup_{\vec{\alpha}'} \sup_{S' \in \text{NE}(\vec{\alpha}')} u_p(S', \vec{\alpha}') = \kappa \cdot u_p(S^*, \vec{\alpha}^*),$$

where $\text{NE}(\vec{\alpha}')$ constitutes the set of actions that establish a Nash equilibrium between the agents under contract $\vec{\alpha}'$. We note that the supremum in the last equation is attained at some optimal contract $\vec{\alpha}^*$ with corresponding equilibrium S^* (see Proposition B.2). We say that a pair of contract and actions is optimal if $\kappa = 1$. Otherwise, $\kappa \in [0, 1)$ captures how close to optimal the solution is.

We remark that our algorithmic results are actually stronger: We find a contract $\vec{\alpha}$ such that any choice of actions S' that constitutes an equilibrium for this $\vec{\alpha}$ achieves the claimed approximation guarantee. Namely, we find a contract $\vec{\alpha}$ such that

$$\inf_{S' \in \text{NE}(\vec{\alpha})} u_p(S', \vec{\alpha}) \geq \kappa \cdot u_p(S^*, \vec{\alpha}^*).$$

Classes of reward functions. We generally assume that the reward function f is normalized so that $f(\emptyset) = 0$, and that it is non-decreasing, i.e., that for any $S \subseteq S' \subseteq A$ it holds that $f(S) \leq f(S')$. We sometimes use the shorthand $f(j) = f(\{j\})$. For an action j and a set of actions $S \subseteq A$, we write $f(j \mid S) = f(S \cup \{j\}) - f(S)$ for the marginal contribution of j to S . Our main interest will be in the following classes of reward functions $f : 2^A \rightarrow [0, 1]$ that come from the hierarchy of complement-free functions [39]:

- Set function f is *additive* if there exist values $v_1, \dots, v_m \in \mathbb{R}_+$ such that $f(S) = \sum_{j \in S} v_j$.
- A set function f is *gross substitutes* if for any two vectors $p, q \in \mathbb{R}_+^m$ and any $S \subseteq A$ such that $f(S) - \sum_{j \in S} p_j \in \arg \max_{S' \subseteq A} (f(S') - \sum_{j \in S'} p_j)$ there is a $T \subseteq A$ such that $f(T) - \sum_{j \in T} q_j \in \arg \max_{T' \subseteq A} (f(T') - \sum_{j \in T'} q_j)$ and $T \supseteq \{j \in S \mid q_j \leq p_j\}$.
- Set function f is *submodular* if for every two sets $S, S' \subseteq A$ with $S \subseteq S'$ and any $j \in A$ it holds that $f(j \mid S) \geq f(j \mid S')$.
- Set function f is *XOS* if there exist a collection of additive functions $\{a_s : 2^A \rightarrow \mathbb{R}_+\}_{s=1, \dots, k}$ such that for each $S \subseteq A$ it holds that $f(S) = \max_{\ell=1, \dots, k} \sum_{j \in S} a_{\ell j}$. Given an XOS function f and a set S there exists an additive function a_ℓ such that $a_\ell(S) = f(S)$ and $a_\ell(T) \leq f(T)$ for all $T \subseteq A$; this function is called the *additive supporting function* of f on S .
- Set function f is *subadditive* if for any two sets $S, S' \subseteq A$, it holds that $f(S) + f(S') \geq f(S \cup S')$.

A well-known fact is that additive $\subseteq$ gross substitutes $\subseteq$ submodular $\subseteq$ XOS $\subseteq$ subadditive, and that all containment relations are strict [39].

We emphasize that we do *not* assume additive separability of f across agents, i.e., we do *not* assume that $f(S) = \sum_i f_i(S_i)$. While additive separability is a natural assumption in combinatorial auctions (where notions such as the welfare of an allocation are defined as the sum of the agents' valuations), it would preclude natural effects between agents and the actions they take in the contracts setting.

Value and demand oracles. As is common in the literature on combinatorial optimization problems with set functions, we consider two types of primitives for accessing the reward function f .

- *Value oracle access:* A value oracle for f is given a set $S \subseteq A$ and returns the value $f(S)$.
- *Demand oracle access:* A demand oracle is given f and a (price) vector $p \in \mathbb{R}_+^m$ and returns a set $S \subseteq A$ that maximizes $f(S) - \sum_{j \in S} p_j$.

Both value and demand oracles are standard in the literature on combinatorial optimization. They also naturally come up in the context of combinatorial contracts. For instance, in the single-agent combinatorial actions case, a demand oracle corresponds to the agent's best response problem [18]. Indeed, there the agent's best response to a contract α is the set $S \subseteq A$ that maximizes $\alpha f(S) - c(S)$, or equivalently, dividing through α and interpreting $f(S)$ as the agent's valuation, the set S that maximizes $f(S) - \sum_{j \in S} p_j$ at prices $p_j = c_j/\alpha$. More generally, in the multi-agent case, for a given contract $\vec{\alpha}$, a demand oracle corresponds to the best set of actions to take, had the principal to take the actions herself [19]. Specifically, Proposition 2.1 shows that, for any contract $\vec{\alpha}$, a demand oracle at prices p_j/α_i returns some equilibrium S .

3 A Constant-Factor Approximation When No Agent Is Large

As a first ingredient to our proof of Main Theorem 1.1, we show how to get a constant-factor approximation to the best contract in which no agent is large.

THEOREM 3.1. *For every submodular success probability f , Algorithm 3 runs in polynomial time using polynomially many value queries, and returns a contract $\vec{\alpha}$ with a guarantee Λ such that in any equilibrium S of $\vec{\alpha}$ the principal's utility is at least Λ . The guarantee Λ satisfies that: There exist constants $\phi, \kappa > 0$ such that for any contract $\vec{\alpha}^*$ and any equilibrium S^* of $\vec{\alpha}^*$ such that $f(S_i^*) \leq \phi \cdot f(S^*)$ for all $i \in [n]$, it holds that $\Lambda \geq \kappa \cdot (1 - \sum_{i \in [n]} \alpha_i^*) f(S^*)$.*

Proof outline. The proof of Theorem 3.1 is constructed step-by-step in the subsequent subsections as detailed below.

In Section 3.1 we define the notion of a *subset-stable* set with respect to a contract. Subset stability is a relaxation of an equilibrium, where the set is stable against removal of actions (any subset thereof) but not against the addition of actions. Subsequently, we establish a highly useful property of subset-stable sets, under submodular reward functions, as specified in the “Doubling Lemma.” This property suggests that, upon doubling the contract to all agents, the reward in *any* equilibrium of the doubled contract will be at least half of that in the subset-stable set.

However, identifying subset-stable sets with high value and relatively small payments remains a challenge due to the exponential number of incentive constraints. To address this, in Section 3.2, we identify a simple sufficient condition for a set to be subset stable with respect to a contract. This condition specifies bundle pricing (which is not necessarily additive) for which a demand set satisfies this condition.

But now a new challenge emerges: answering this demand query with respect to the corresponding bundle prices is hard. To overcome this challenge, in Section 3.3 we devise an algorithm that gives an approximate demand set with respect to the bundle prices.

Finally, in Section 3.4, we show how to transform the approximate demand set into a subset-stable set (with respect to a contract that pays at most a quarter of the reward) that approximates the optimal contract with no large agent.

3.1 Subset Stability and the Doubling Lemma While it is typically insufficient to ensure stability against “subset deviations” (namely, a deviation to a strict subset of the current set of actions), we show that ensuring such stability is sufficient in some precise approximate sense. Before formalizing this claim, we introduce the notion of subset stability.

DEFINITION 3.1. (SUBSET STABILITY) *A set of actions S is subset stable with respect to a contract $\vec{\alpha}$ if for all agents $i \in [n]$ and all $S'_i \subseteq S_i$ it holds that*

$$\alpha_i f(S_i, S_{-i}) - c(S_i) \geq \alpha_i f(S'_i, S_{-i}) - c(S'_i).$$

The following key lemma formalizes the argument that subset stability suffices approximately. This lemma will serve us more than once throughout the paper.

LEMMA 3.1. (THE DOUBLING LEMMA) *Suppose f is submodular. Let $\epsilon > 0$ and let $\vec{\epsilon} = (\epsilon, \dots, \epsilon) \in \mathbb{R}_+^n$. Let S be a subset-stable action set with respect to a contract $\vec{\alpha}$. Then any equilibrium $S^\dagger$ with respect to $2\vec{\alpha} + \vec{\epsilon}$ fulfills $f(S^\dagger) \geq \frac{1}{2}f(S)$.*

We remark that Lemma 3.1 is quite strong. In particular, it holds for *any* subset-stable action set and *any* equilibrium with respect to $2\vec{\alpha} + \vec{\epsilon}$. Indeed, this lemma holds with respect to submodular functions f , but fails for XOS functions, where there are instances in which any contract admits some bad equilibrium (see Proposition B.1). We proceed with the proof of Lemma 3.1.

Proof. [of Lemma 3.1] As $S^\dagger$ is an equilibrium with respect to $2\vec{\alpha} + \vec{\epsilon}$, agent i weakly prefers $S_i^\dagger$ over $S_i^\dagger \cup S_i$. That is,

$$(2\alpha_i + \epsilon)f(S^\dagger) - c(S_i^\dagger) \geq (2\alpha_i + \epsilon)f(S_i^\dagger \cup S_i, S_{-i}^\dagger) - c(S_i^\dagger \cup S_i)$$

and therefore

$$(2\alpha_i + \epsilon) \left(f(S_i^\dagger \cup S_i, S_{-i}^\dagger) - f(S^\dagger) \right) \leq c(S_i^\dagger \cup S_i) - c(S_i^\dagger) = c(S_i \setminus S_i^\dagger).$$

Furthermore, since S is subset stable with respect to $\vec{\alpha}$, agent i weakly prefers S_i over $S_i^\dagger \cap S_i \subseteq S_i$. So, we have

$$\alpha_i f(S) - c(S_i) \geq \alpha_i f(S_i^\dagger \cap S_i, S_{-i}) - c(S_i^\dagger \cap S_i).$$

This implies

$$\alpha_i \left(f(S) - f(S_i^\dagger \cap S_i, S_{-i}) \right) \geq c(S_i) - c(S_i^\dagger \cap S_i) = c(S_i \setminus S_i^\dagger).$$

In combination, we obtain

$$(3.1) \quad f(S_i^\dagger \cup S_i, S_{-i}^\dagger) - f(S^\dagger) \leq \frac{\alpha_i}{2\alpha_i + \epsilon} (f(S) - f(S_i^\dagger \cap S_i, S_{-i})) \leq \frac{1}{2} (f(S) - f(S_i^\dagger \cap S_i, S_{-i})),$$

where we use that the expression is non-negative by monotonicity of f .

Let $S_i^j = S_i^\dagger \cup S_i$ if $i \leq j$ and $S_i^j = S_i^\dagger$ otherwise. By this definition

$$f(S) \leq f(S^n) = f(S^0) + \sum_{j=1}^n (f(S^j) - f(S^{j-1})).$$

Note that $S^0 = S^\dagger$. Furthermore, by submodularity of f , we have

$$f(S^i) - f(S^{i-1}) = f(S_i \setminus S_i^\dagger \mid S^{i-1}) \leq f(S_i \setminus S_i^\dagger \mid S^\dagger) = f(S_i^\dagger \cup S_i, S_{-i}^\dagger) - f(S^\dagger).$$

Therefore, we have

$$\begin{aligned} f(S) &\leq f(S^\dagger) + \sum_{i=1}^n (f(S_i^\dagger \cup S_i, S_{-i}^\dagger) - f(S^\dagger)) \leq f(S^\dagger) + \sum_{i=1}^n \frac{1}{2} (f(S) - f(S_i^\dagger \cap S_i, S_{-i})) \\ &= f(S^\dagger) + \frac{1}{2} \cdot \sum_{i=1}^n f(S_i \setminus S_i^\dagger \mid S_i^\dagger \cap S_i, S_{-i}) \leq f(S^\dagger) + \frac{1}{2} \cdot \sum_{i=1}^n f(S_i \setminus S_i^\dagger \mid S_{<i} \setminus S^\dagger) \\ &= f(S^\dagger) + \frac{1}{2} f(S \setminus S^\dagger), \end{aligned}$$

where in the second inequality we used Equation (3.1), and in the third inequality we use submodularity. So, we have $f(S) \leq 2f(S^\dagger)$. $\square$

3.2 A Sufficient Condition for Subset Stability The following lemma defines a sufficient condition for a set to be subset stable for some contract. This is the key ingredient that drives our algorithm for finding subset stable sets with a relatively high value for which the corresponding contract does not pay more than half of the utility.

LEMMA 3.2. *Suppose f is submodular. Fix $\gamma > 0$ and let S be a set such that for all i and all $j \in S_i$, we have $f(j \mid S \setminus \{j\}) \geq \gamma \left(\sqrt{c(S_i)} - \sqrt{c(S_i \setminus \{j\})} \right)$. Then, S is subset stable under the contract $\vec{\alpha}$ defined by $\alpha_i = \frac{4}{\gamma} \sqrt{c(S_i)}$.*

The condition in Lemma 3.2 has the following natural economic interpretation: For “bundle prices” $p(S) = \gamma \sum_{i \in [n]} \sqrt{c(S_i)}$, if the (quasi-linear) “utility” $f(S) - p(S)$ cannot be improved upon by dropping any single action then the set is subset stable under contract $\vec{\alpha}$ (as defined in the lemma).

One desirable property of this condition is that, unlike subset stability, it is easy to verify (we just need to check that we don’t want to drop individual actions). Our algorithms in Sections 3.3 and 3.4 are inspired by this economic interpretation.

Proof. [of Lemma 3.2] Let $S'_i \subseteq S_i$, and let $X = S_i \setminus S'_i$. By submodularity, we have

$$(3.2) \quad f(S_i \mid S_{-i}) - f(S'_i \mid S_{-i}) \geq \sum_{j \in X} f(j \mid S \setminus \{j\}).$$

Furthermore, each such term satisfies

$$(3.3) \quad f(j \mid S \setminus \{j\}) \geq \gamma (\sqrt{c(S_i)} - \sqrt{c(S_i \setminus \{j\})}) = \gamma \frac{c(S_i) - c(S_i \setminus \{j\})}{\sqrt{c(S_i)} + \sqrt{c(S_i \setminus \{j\})}} \geq \frac{\gamma c(\{j\})}{2\sqrt{c(S_i)}}.$$

By abbreviated multiplication we have that

$$(3.4) \quad c(S_i) - c(S'_i) = (\sqrt{c(S_i)} - \sqrt{c(S'_i)})(\sqrt{c(S_i)} + \sqrt{c(S'_i)}) \geq (\sqrt{c(S_i)} - \sqrt{c(S'_i)})\sqrt{c(S_i)}.$$

Combining Equations (3.2), (3.3), and (3.4) we get:

$$\begin{aligned} f(S_i \mid S_{-i}) - f(S'_i \mid S_{-i}) &\stackrel{(3.2)}{\geq} \sum_{j \in X} f(j \mid S \setminus \{j\}) \stackrel{(3.3)}{\geq} \sum_{j \in X} \frac{\gamma c(\{j\})}{2\sqrt{c(S_i)}} \\ &= \frac{\gamma c(X)}{2\sqrt{c(S_i)}} = \frac{\gamma}{2\sqrt{c(S_i)}} (c(S_i) - c(S'_i)) \stackrel{(3.4)}{\geq} \frac{\gamma}{2} (\sqrt{c(S_i)} - \sqrt{c(S'_i)}). \end{aligned}$$

This implies that

$$\frac{c(S_i) - c(S'_i)}{f(S_i \mid S_{-i}) - f(S'_i \mid S_{-i})} \leq \frac{c(S_i) - c(S'_i)}{\frac{\gamma}{2} (\sqrt{c(S_i)} - \sqrt{c(S'_i)})} = \frac{2}{\gamma} \left(\sqrt{c(S_i)} + \sqrt{c(S'_i)} \right) \leq \frac{4}{\gamma} \sqrt{c(S_i)} = \alpha_i,$$

where we use that $c(S'_i) \leq c(S_i)$ because $S'_i \subseteq S_i$.

By rearrangement of the last inequality we get that $\alpha_i f(S) - c(S_i) \geq \alpha_i f(S'_i, S_{-i}) - c(S'_i)$, which concludes the proof. $\square$

3.3 Approximate Demand Oracles for Bundle Prices Lemma 3.2 states that if $f(S) - p(S)$ cannot be increased by dropping actions, the set S is subset stable under the contract $\vec{\alpha}$ defined there. Clearly, a set S that maximizes $f(S) - p(S)$ fulfills this property. Motivated by this, in this section, we seek to approximately maximize $f(S) - p(S)$. This approximation algorithm will then be used as a subroutine in our final algorithm to compute a contract.

To keep notation clean and abstract away from our application, we consider the following problem. Let A be any set partitioned into $A_1, \dots, A_n$. Furthermore, let $h: 2^A \rightarrow \mathbb{R}$ be any monotone submodular function and let $p: 2^A \rightarrow \mathbb{R}$ be defined by $p(S) = \sum_{i \in [n]} \sqrt{\sum_{j \in S_i} q_j}$, where $q_1, \dots, q_m \geq 0$ and $S_i = S \cap A_i$. Our goal is to find a set S that maximizes $h(S) - p(S)$.

3.3.1 Single Agent To simplify our optimization problem, we first consider the case that $n = 1$. That is, our goal is now to maximize $h(S) - \sqrt{\sum_{j \in S} q_j}$. For this problem, we propose Algorithm 1. The idea behind this algorithm is as follows. Suppose we knew the value $h(S^*)$ for the optimal solution S^* . Then define additive prices $\tilde{p}_j \approx \frac{q_j}{h(S^*)}$ and compute an approximate demand set with respect to these additive prices.

Algorithm 1 Approximate demand oracle for bundle prices for a single agent

Input: Costs $q_1, \dots, q_m \in \mathbb{R}_{\geq 0}$, and value oracle access to a submodular function $h : 2^A \rightarrow \mathbb{R}_{\geq 0}$

Output: A set S

```

1: for  $j \in A$  and  $k \in \{1, \dots, \lceil \log m \rceil\}$  do
2:   Let  $x = 2^k \cdot h(j)$ 
3:    $S^x = \emptyset$ 
4:   Set  $\tilde{p}_{j'} = \frac{18q_{j'}}{x}$  for all  $j' \in M$ 
5:   while there is  $j' \in M \setminus S^x$  such that  $h(j' \mid S^x) - \tilde{p}_{j'} \geq 0$  do
6:     Add to  $S^x$  the element  $j'$  that maximizes  $h(j' \mid S^x) - \tilde{p}_{j'}$ 
7:   end while
8: end for
9: return  $S^x$  for which  $h(S^x) - \sqrt{q(S^x)}$  is maximized

```

THEOREM 3.2. For $\rho = \frac{1}{6}$, Algorithm 1 computes a set S satisfying

$$h(S) - \sqrt{\sum_{j \in S} q_j} \geq \max_{S^*} \left(\rho \cdot h(S^*) - \sqrt{\sum_{j \in S^*} q_j} \right).$$

Proof. Let S^* maximize $\rho \cdot h(S^*) - \sqrt{\sum_{j \in S^*} q_j}$. If $S^* = \emptyset$ then the guarantee is trivial, since the algorithm returns a set with a non-negative value of $h(S) - \sqrt{\sum_{j \in S} q_j}$. Otherwise, let $j^* = \arg \max_{j \in S^*} h(j)$ note that $h(j^*) \leq h(S^*) \leq m \cdot h(j^*)$, where the second inequality follows by subadditivity. So, there is an execution of the algorithm in which $x/2 \leq h(S^*) \leq x$ since when the algorithm considers $j = j^*$, then for $k = 1$ it holds that $x/2 \leq h(j^*)$ while for $k = \lceil \log m \rceil$ it holds that $x \geq m \cdot h(j^*)$.

OBSERVATION 3.1. The algorithm computes a set S such that $h(S) \geq \sum_{j \in S} \tilde{p}_j$.

LEMMA 3.3. The algorithm computes a set S such that $h(S) \geq \frac{1}{2}h(S^*)$.

Proof. Upon termination it holds that $h(j \mid S) - \tilde{p}_j \leq 0$ for all $j \in M$.

So

$$\begin{aligned}
0 &\geq \max_{j \in M} (h(j \mid S) - \tilde{p}_j) \geq \frac{1}{|S^*|} \sum_{j \in S^*} (h(j \mid S) - \tilde{p}_j) \\
&\geq \frac{1}{|S^*|} \left(h(S^* \cup S) - h(S) - \sum_{j \in S^*} \tilde{p}_j \right) \geq \frac{1}{|S^*|} \left(h(S^*) - h(S) - \sum_{j \in S^*} \tilde{p}_j \right),
\end{aligned}$$

where the third inequality is by submodularity of h , and the fourth inequality is by monotonicity of h . Combined with the fact that $|S^*| \geq 1$, this implies that $h(S) \geq h(S^*) - \sum_{j \in S^*} \tilde{p}_j$. Furthermore

$$\sum_{j \in S^*} \tilde{p}_j = \sum_{j \in S^*} \frac{18q_j}{x} \leq \frac{18(\rho \cdot h(S^*))^2}{x} \leq \frac{1}{2}h(S^*),$$

where the first inequality is since $\rho \cdot h(S^*) - \sqrt{\sum_{j \in S^*} q_j} \geq \rho \cdot h(\emptyset) - \sqrt{\sum_{j \in \emptyset} q_j} = 0$, and the second inequality is since $x \geq h(S^*)$ and by the value of ρ . $\square$

By combining (1) Observation 3.1, (2) that $x \leq 2h(S^*)$, and (3) Lemma 3.3 we get that:

$$\sum_{j \in S} q_j = \frac{x}{18} \sum_{j \in S} \tilde{p}_j \leq \frac{2h(S^*)}{18} h(S) \leq \frac{4}{18} h(S)^2,$$

which by rearrangement gives us that $\sqrt{\sum_{j \in S} q_j} \leq \frac{\sqrt{2}}{3} h(S)$. Therefore, it holds that

$$h(S) - \sqrt{\sum_{j \in S} q_j} \geq \frac{3 - \sqrt{2}}{3} h(S) \geq \rho \cdot h(S^*) \geq \rho \cdot h(S^*) - \sqrt{\sum_{j \in S^*} q_j},$$

which concludes the proof. $\square$

3.3.2 Multiple Agents We now turn to the actual problem of maximizing $h(S) - \sum_{i \in [n]} \sqrt{\sum_{j \in S_i} q_j}$. Algorithm 2 repeatedly calls Algorithm 1. In these calls, it sets h' to be the marginal increases and modifies $q_1, \dots, q_m$ by setting $\tilde{q}_j = 4q_j$.

Algorithm 2 Approximate demand oracle for bundle prices for multiple agents

Input: Costs $q_1, \dots, q_m \in \mathbb{R}_{\geq 0}$, and value oracle access to a submodular function $h : 2^A \rightarrow \mathbb{R}_{\geq 0}$

Output: A set S

- 1: Let $\tilde{q}_j = 4q_j$ for every $j \in A$
 - 2: **for** $t = 1, \dots, n$ **do**
 - 3: **for** $i = 1, \dots, n$ **do**
 - 4: Let $h' : A_i \rightarrow \mathbb{R}_{\geq 0}$ be such that $h'(S) = h(S^{(t,i)} \mid S^{(1)} \cup \dots \cup S^{(t-1)})$
 - ▷ Note that a value query to h' can be computed using two value queries to h
 - ▷ If h is monotone and submodular then h' is monotone, submodular, and normalized
 - 5: Call Algorithm 1 on $\tilde{q}_1, \dots, \tilde{q}_m$ and h' to obtain $S^{(t,i)} \subseteq A_i$
 - 6: **end for**
 - 7: Let $S^{(t)}$ be the set that maximizes $h(S^{(t,i)} \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \sqrt{\sum_{j \in S^{(t,i)}} \tilde{q}_j}$ over all i
 - 8: **end for**
 - 9: **return** $S = S^{(1)} \cup \dots \cup S^{(n)}$.
-

THEOREM 3.3. For the parameter ρ of Theorem 3.2, the set S returned by Algorithm 2 fulfills

$$h(S) - \sum_{i \in [n]} \sqrt{\sum_{j \in S_i} q_j} \geq \max_{S^*} \left(\frac{\rho}{2} h(S^*) - \sum_{i \in [n]} \sqrt{\sum_{j \in S_i^*} q_j} \right).$$

Proof. Let $(S_1^*, \dots, S_n^*)$ be any sets with $S_i^* \subseteq A_i$.

Write $\tilde{p}(X) = \sum_{i \in [n]} \sqrt{\sum_{j \in X_i} \tilde{q}_j}$. Note that by this definition $\tilde{p}(S^*) = \sum_{i \in [n]} \tilde{p}(S_i^*)$ because $(S_1^*, \dots, S_n^*)$ are disjoint and each belongs to a different A_i . Furthermore $\tilde{p}(S) \leq \sum_{t=1}^n \tilde{p}(S^{(t)})$ because $\tilde{p}$ is subadditive.

For every $t = 1, \dots, n$, by Theorem 3.2 we have

$$\begin{aligned}
(3.5) \quad & h(S^{(t)} \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \tilde{p}(S^{(t)}) \\
& \geq \max_{i \in [n]} \left(\rho h(S_i^* \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \tilde{p}(S_i^*) \right) \\
& \geq \frac{1}{n} \sum_{i \in [n]} \left(\rho h(S_i^* \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \tilde{p}(S_i^*) \right) \\
& \geq \frac{1}{n} \sum_{i \in [n]} \left(\rho h(S_i^* \mid S^{(1)} \cup \dots \cup S^{(t-1)} \cup S_1^* \cup \dots \cup S_{i-1}^*) - \tilde{p}(S_i^*) \right) \\
& = \frac{1}{n} \left(\rho h(S_1^* \cup \dots \cup S_n^* \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \tilde{p}(S^*) \right) \\
& \geq \frac{1}{n} \left(\rho \left(h(S_1^* \cup \dots \cup S_n^*) - h(S^{(1)} \cup \dots \cup S^{(t-1)}) \right) - \tilde{p}(S^*) \right) \\
& \geq \frac{1}{n} \left(\rho (h(S^*) - h(S)) - \tilde{p}(S^*) \right).
\end{aligned}$$

Thus,

$$\begin{aligned}
h(S) - \tilde{p}(S) & \geq h(S) - \sum_{t=1}^n \tilde{p}(S^{(t)}) \\
& = \sum_{t=1}^n \left(h(S^{(t)} \mid S^{(1)} \cup \dots \cup S^{(t-1)}) - \tilde{p}(S^{(t)}) \right) \\
& \stackrel{(3.5)}{\geq} \sum_{t=1}^n \frac{1}{n} (\rho (h(S^*) - h(S)) - \tilde{p}(S^*)) \\
& = \rho h(S^*) - \rho h(S) - \tilde{p}(S^*).
\end{aligned}$$

Therefore, since $\rho \leq 1$, we get that

$$2h(S) - \tilde{p}(S) \geq (1 + \rho)h(S) - \tilde{p}(S) \geq \rho h(S^*) - \tilde{p}(S^*)$$

and thus $h(S) - \frac{1}{2}\tilde{p}(S) \geq \frac{\rho}{2}h(S^*) - \frac{1}{2}\tilde{p}(S^*)$, which by plugging the definitions of $\tilde{p}, \tilde{q}$ concludes the proof. $\square$

3.4 Approximation Algorithm Under No Large Agent We are now ready to present our algorithm (Algorithm 3) that finds an approximate optimal contract among all contracts under the assumption of no large agents.

Recall that in this section $\vec{\alpha}^*, S^*$ is the best contract and the corresponding best equilibrium for which $f(S_i^*) \leq \phi f(S^*)$ for all $i \in [n]$. We assume that $f(S^*) > 0$ (as otherwise the instance is redundant and our algorithm always returns a contract with non-negative utility). Furthermore, $\sum_{i \in [n]} \alpha_i^* \leq 1$ because otherwise the principal utility would be negative.

Algorithm 3 first guesses an estimate of $f(S^*)$, which is captured by ξ . Given this estimate, the algorithm defines a new instance with a reward function that is capped at ξ , and applies Algorithm 2 for this modified instance which returns a set S . From this returned set, the algorithm removes actions with negative marginal contribution with respect to the bundle prices defined by $p(\cdot)$. Among all such created sets, the algorithm returns the one with the maximum value of $f(\cdot)$ (over all choices of ξ).

We analyze the guarantees of Algorithm 3 with parameter ρ set as in Theorem 3.2. We first claim that the algorithm is well defined. For this we note that since $f(S^*) > 0$, there exists $j^* \in A$ with $f(j^*) > 0$, which means that γ^* is well defined (as it is an argmax of a non-empty set). We also note that a value query to $\hat{f}$ can be computed using a value query to f in order to call Algorithm 2 in Step 5, and that the function $\hat{f}$ is monotone, submodular, and normalized since f is monotone, submodular, and normalized, which satisfies the requirements of Algorithm 2.

We now observe that there exists an iteration for which ξ is close to $\frac{\rho^2}{512}f(S^*)$

Algorithm 3 Approximate optimal contract under no large agent

Parameters: $\rho \in [0, 1]$, $\epsilon \in (0, \frac{1}{4n})$
Input: Costs $c_1, \dots, c_m \in \mathbb{R}_{\geq 0}$, and value oracle access to a submodular function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$
Output: A contract $\vec{\alpha}$ with a guarantee of principal's utility of Λ for any equilibrium of $\vec{\alpha}$

```

1: for  $j^* \in A$  with  $f(j^*) > 0$  and  $t = 0, \dots, \lceil \log m \rceil$  do
2:   Let  $\xi = \frac{\rho^2}{512} \cdot f(j^*) \cdot 2^t$  and  $\gamma = \sqrt{32\xi}$ 
3:   Let  $\hat{f} : 2^A \rightarrow \mathbb{R}_{\geq 0}$  be such that  $\hat{f}(S) = \min\{f(S), \xi\}$ 
4:   Let  $p : 2^A \rightarrow \mathbb{R}_{\geq 0}$  be defined by  $p(S) = \gamma \sum_{i \in [n]} \sqrt{\sum_{j \in S_i} c_j}$ 
5:   Apply Algorithm 2 on  $h = \hat{f}$  and  $q_j = \gamma^2 c_j$  to maximize  $\hat{f}(S) - p(S)$ , let  $S^\gamma$  be its output
6:   while exist  $j \in S$  such that  $\hat{f}(j \mid S^\gamma \setminus \{j\}) < p(S^\gamma) - p(S^\gamma \setminus \{j\})$  do
7:      $S^\gamma = S^\gamma \setminus \{j\}$ 
8:   end while
9:   Let  $\alpha_i^\gamma = \frac{4}{\gamma} \sqrt{c(\hat{S}_i)}$ 
10: end for
11: Let  $\gamma^* = \arg \max_\gamma f(S^\gamma)$ 
12: return  $\vec{\alpha} = 2\vec{\alpha}^{\gamma^*} + \vec{\epsilon}$ ,  $\Lambda = \frac{f(S^{\gamma^*})}{8}$ 

```

OBSERVATION 3.2. *There exist $j^* \in A$, and $t \in \{0, \dots, \lceil \log m \rceil\}$ such that $\frac{\rho^2}{1024} f(S^*) \leq \xi \leq \frac{\rho^2}{512} f(S^*)$.*

Proof. Since $f(S^*) > 0$ then $S^* \neq \emptyset$ and for the iteration with $j^* \in \arg \max_{j \in S^*} f(j)$ it holds for $t = 0$ then $\xi = \frac{\rho^2}{512} f(j^*) \leq \frac{\rho^2}{512} f(S^*)$ and for $t = \lceil \log m \rceil$ it holds that $\xi = \frac{\rho^2}{512} f(j^*) \cdot 2^{\lceil \log m \rceil} \geq \frac{\rho^2}{512} f(j^*) \cdot m \geq \frac{\rho^2}{512} f(S^*)$. Thus there must be a value of t is each time multiplied by a factor of 2, ξ must be in the range for one iteration. $\square$

Let γ^* be its corresponding value calculated in Step 11, let $\hat{S} = S^{\gamma^*}$, let $\xi^* = \frac{\gamma^{*2}}{32}$ be its corresponding value of ξ , and let $\vec{\alpha} = (\alpha_1, \dots, \alpha_n) = (\alpha_1^{\gamma^*}, \dots, \alpha_n^{\gamma^*})$ be their corresponding values at the termination of Algorithm 3. We first bound the sum of α s.

LEMMA 3.4. *It holds that $\sum_{i \in [n]} \alpha_i \leq \frac{1}{4}$.*

Proof. For every $i \in [n]$ it holds that

$$\begin{aligned}
 \hat{f}(\hat{S}_i \mid \hat{S}_{-i}) &\geq \sum_{j \in \hat{S}_i} \hat{f}(j \mid \hat{S} \setminus \{j\}) \geq \sum_{j \in \hat{S}_i} \gamma^* (\sqrt{c(\hat{S}_i)} - \sqrt{c(\hat{S}_i \setminus \{j\})}) \\
 (3.6) \quad &= \gamma^* \sum_{j \in \hat{S}_i} \frac{c(\hat{S}_i) - c(\hat{S}_i \setminus \{j\})}{\sqrt{c(\hat{S}_i)} + \sqrt{c(\hat{S}_i \setminus \{j\})}} \geq \gamma^* \sum_{j \in \hat{S}_i} \frac{c(j)}{2\sqrt{c(\hat{S}_i)}} = \frac{\gamma^*}{2} \sqrt{c(\hat{S}_i)},
 \end{aligned}$$

where the first inequality is by submodularity, the second inequality by Step 6 of the algorithm, and the third inequality is by additivity and monotonicity of the function c .

Thus,

$$\sum_i \alpha_i = \sum_i \frac{4}{\gamma^*} \sqrt{c(\hat{S}_i)} \leq \sum_i \frac{4}{\gamma^*} \cdot \frac{2}{\gamma^*} \hat{f}(\hat{S}_i \mid \hat{S}_{-i}) \leq \frac{8}{\gamma^{*2}} \hat{f}(\hat{S}) \leq \frac{8}{\gamma^{*2}} \xi^* = \frac{1}{4},$$

where the first inequality is by Inequality (3.6), the second inequality is by submodularity, and the third inequality is by the definition of $\hat{f}$. $\square$

We next show that $\hat{S}$ is subset stable.

LEMMA 3.5. *The set $\hat{S}$ is subset stable with respect to contract $\vec{\alpha}$.*

Proof. For all agents $i \in [n]$ and all actions $j \in A_i$ it holds that

$$(3.7) \quad f(j \mid \hat{S} \setminus \{j\}) \geq \hat{f}(j \mid \hat{S} \setminus \{j\}) \geq p(\hat{S}_i) - p(\hat{S}_i \setminus \{j\}) = \gamma^* \cdot \left(\sqrt{c(\hat{S}_i)} - \sqrt{c(\hat{S}_i \setminus \{j\})} \right),$$

where the first inequality is since capping a function by a constant can only decrease the marginal utility, and the second inequality is ensured by Step 6. Thus, the lemma follows by Lemma 3.2, Inequality (3.7) and the definition of α_i . $\square$

We next bound the value of $f(\hat{S})$.

LEMMA 3.6. *If S^* is an equilibrium for $\vec{\alpha}^*$ fulfilling $\sum_{i \in [n]} \alpha_i^* \leq 1$ and $f(S_i^*) \leq \phi f(S^*)$ for all $i \in [n]$, then $f(\hat{S}) \geq \left(\frac{\rho^3}{6144} - \frac{\rho \cdot \phi}{2} \right) f(S^*)$.*

Proof. If $\phi \geq \frac{\rho^2}{3072}$ or $f(S^*) = 0$ then the claim holds trivially since $\left(\frac{\rho^3}{6144} - \frac{\rho \cdot \phi}{2} \right) f(S^*) \leq 0$. Since S^* is an equilibrium, no agent i can increase his utility by dropping all their actions. That is $\alpha_i^* f(S^*) - c(S_i^*) \geq \alpha_i^* f(S^* \setminus S_i^*)$. So, it holds that $\alpha_i^* \geq \frac{c(S_i^*)}{f(S_i^* \mid S^* \setminus S_i^*)}$. Furthermore, $\sum_{i \in [n]} \alpha_i^* \leq 1$. Using the Cauchy–Schwarz inequality $(\sum_i x_i y_i)^2 \leq \sum_i x_i^2 \sum_i y_i^2$ with $x_i = \sqrt{\frac{c(S_i^*)}{f(S_i^* \mid S^* \setminus S_i^*)}}$ and $y_i = \sqrt{f(S_i^* \mid S^* \setminus S_i^*)}$, we get

$$\left(\sum_i \sqrt{c(S_i^*)} \right)^2 \leq \sum_i \frac{c(S_i^*)}{f(S_i^* \mid S^* \setminus S_i^*)} \sum_i f(S_i^* \mid S^* \setminus S_i^*) \leq \sum_i \alpha_i^* f(S^*) \leq f(S^*),$$

where the second inequality is by submodularity of f . Therefore $\sum_{i \in [n]} \sqrt{c(S_i^*)} \leq \sqrt{f(S^*)}$.

By Observation 3.2 we know that there is iterations of j^*, t of Algorithm 3 for which their corresponding $\xi \in [\frac{\rho^2}{1024} f(S^*), \frac{\rho^2}{512} f(S^*)]$. We denote this value of ξ by ξ^{**} , and let γ^{**} be its corresponding value of γ (i.e., $\gamma^{**} = \sqrt{32\xi^{**}}$). For γ^{**} , let S^1, S^2 be the values of $S^{\gamma^{**}}$ before and after Steps 6–8 respectively.

Let $B_1, \dots, B_\ell \subseteq [n]$ be an arbitrary partition of the set of agents $[n]$ such that $\xi^{**} - \max_{i \in [n]} f(S_i^*) \leq f(\bigcup_{i \in B_k} S_i^*) \leq \xi^{**}$ for all $k < \ell$, and $f(\bigcup_{i \in B_k} S_i^*) \leq \xi^{**}$ for $k = \ell$. Such a partition exists and can be created by bundling S_i^* greedily until each bundle has a value in the desired range. This greedy process is well defined since $\xi^{**} > \max_i f(S_i^*)$ which holds since $\phi < \frac{\rho^2}{3072}$.

Note that $\ell \geq \frac{f(S^*)}{\xi^{**}}$ because otherwise by subadditivity

$$f(S^*) = f\left(\bigcup_{k=1}^{\ell} \bigcup_{i \in B_k} S_i^*\right) \leq \sum_{k=1}^{\ell} f\left(\bigcup_{i \in B_k} S_i^*\right) \leq \ell \xi^{**} < f(S^*).$$

As $B_1, \dots, B_\ell$ partition $[n]$, it holds that $\sum_{k < \ell} \sum_{i \in B_k} \sqrt{c(S_i^*)} \leq \sum_{i=1}^n \sqrt{c(S_i^*)} \leq \sqrt{f(S^*)}$, and there must exist $k < \ell$ such that $\sum_{i \in B_k} \sqrt{c(S_i^*)} \leq \frac{1}{\ell-1} \sqrt{f(S^*)}$. Let $S^{**} = \bigcup_{i \in B_k} S_i^*$. Note that

$$(3.8) \quad \hat{f}(S^{**}) = f(S^{**}) \geq \xi^{**} - \max_{i \in [n]} f(S_i^*),$$

and

$$(3.9) \quad \sum_{i \in [n]} p(S_i^{**}) \leq \frac{\gamma}{\ell-1} \sqrt{f(S^*)} \leq \frac{\rho \xi^{**}}{3}.$$

Therefore, it holds that

$$\begin{aligned} f(\hat{S}) &\geq f(S^2) \geq \hat{f}(S^2) \geq \hat{f}(S^2) - \sum_{i \in [n]} p(S_i^2) \geq \hat{f}(S^1) - \sum_{i \in [n]} p(S_i^1) \\ &= \hat{f}(S^1) - \sum_{i \in [n]} \gamma \cdot \sqrt{c(S_i^1)} \geq \frac{\rho}{2} \cdot \hat{f}(S^{**}) - \sum_{i \in [n]} \gamma \cdot \sqrt{c(S_i^{**})} \\ &= \frac{\rho}{2} \cdot \hat{f}(S^{**}) - \sum_{i \in [n]} p(S_i^{**}) \geq \frac{\rho \xi^{**}}{6} - \frac{\rho}{2} \max_{i \in [n]} f(S_i^*) \geq \left(\frac{\rho^3}{6144} - \frac{\rho \cdot \phi}{2} \right) f(S^*), \end{aligned}$$

where the fourth inequality is by Theorem 3.3, and the last inequality is by Equations (3.8) and (3.9). $\square$

We are now ready to prove the main theorem of this section.

Proof. [of Theorem 3.1] The algorithm clearly runs in polynomial time using polynomially many value queries. By Lemma 3.4 we know that the sum of alphas at the termination is at most $\frac{1}{4}$, which implies that under contract $2\vec{\alpha} + \vec{\epsilon}$ the principal pays at most $2 \cdot \frac{1}{4} + n \cdot \epsilon \leq \frac{3}{4}$. Let S be some equilibrium with respect to contract $2\vec{\alpha} + \vec{\epsilon}$ then by Lemma 3.5 and Lemma 3.1, we get that $f(S) \geq \frac{1}{2} \cdot f(\hat{S})$, which means that the principal's utility is

$$u_p(S, 2\vec{\alpha} + \vec{\epsilon}) \geq (1 - \sum_i \alpha_i) f(S) \geq \frac{1}{8} f(\hat{S}) = \Lambda.$$

By Lemma 3.6, it holds that $\Lambda = \frac{1}{8} f(\hat{S}) \geq \frac{1}{8} \left(\frac{\rho^3}{6144} - \frac{\rho \cdot \phi}{2} \right) f(S^*)$, which concludes the proof. $\square$

4 Approximating the Single Agent Benchmark

In this section, as our second ingredient for Main Theorem 1.1, we show how to obtain a constant factor approximation against the best single-agent contract.

THEOREM 4.1. *For every submodular success probability f , Algorithm 5 runs in polynomial time using polynomially many value and demand queries, and returns a contract $\vec{\alpha}$ with a guarantee Λ such that in any equilibrium S of $\vec{\alpha}$ the principal's utility is at least Λ . The guarantee Λ satisfies that: For any contract $\vec{\alpha}^*$ and any equilibrium S^* of $\vec{\alpha}^*$ such that $\alpha_i^* = 0$ and $S_i^* = \emptyset$ for all but one $i \in [n]$, it holds that $\Lambda \geq \frac{1}{24} \cdot (1 - \sum_{i \in [n]} \alpha_i^*) f(S^*)$.*

In Section 4.1 we give a FPTAS for the single-agent problem for general f , with value and demand oracle access. This result is of independent interest. Previously, only a weakly poly-time algorithm was known [18].

Then, in Section 4.2, we extend this result to a robust approximation in the multi-agent case with submodular f ; namely, a contract for which *every* induced equilibrium obtains the desired approximation. The challenge arises from the existence of equilibria in which other agents take actions, even when their α_i is zero, which may in turn affect other agents' actions. Our argument relies on the Doubling Lemma (Lemma 3.1).

4.1 FPTAS for a Single Agent Consider a single-agent instance defined by a set A of m actions, a success probability function $f : 2^A \rightarrow [0, 1]$, and additive costs c . Let α^*, S^* be the optimal contract and the optimal set of actions.

Below we present an FPTAS for the optimal contract problem with general rewards f , with value and demand oracle access. Previously, only a weakly-polynomial time FPTAS was known [18]. Our result improves on this by showing that an arbitrarily good approximation to the optimal contract can be found in poly-time, independent of the representation size of the input.

Our result is best possible in two ways: First, it is known that computing an optimal contract is NP-hard for submodular f [18]. Second, it was recently shown that computing an optimal contract for submodular f requires exponentially many demand oracle calls [20].

THEOREM 4.2. (FPTAS FOR SINGLE-AGENT PROBLEM) *Consider the single-agent problem with general f . Then Algorithm 4 gives a $(1 - \epsilon)$ -approximation to the optimal principal utility with $O\left(\frac{m^2}{\epsilon}\right)$ many value and demand queries.*

The main challenge of the single agent case is that the optimal α^* could be very close to 1. Indeed, if we knew that α^* was bounded away from 1, we could find a close to optimal contract through a fine-enough discretization of the contract space. To bound α^* , we use that, by a reduction to the non-combinatorial model of [25], the gap between the maximum welfare and the principal's utility under the optimal contract is at most 2^m .

OBSERVATION 4.1. ([25]) *For every reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ over $|A| = m$ actions and a single agent with costs $c_1, \dots, c_m$, there exists a contract α that guarantees a utility for the principal of at least $\frac{OPT}{2^m}$, where $OPT = \max_S (f(S) - c(S))$.*

Using this observation, we now show our key lemma, which sets lower and upper bounds on α^* .

LEMMA 4.1. *Consider the single-agent problem with reward function f . Let α^* , S^* be the optimal contract and the optimal set of actions, respectively, let $j^* \in \arg \max_{j \in S^*} c_j$, and let $OPT = \max_S (f(S) - c(S)) > 0$. Then we have $\alpha_{\min} \leq \alpha^* \leq \alpha_{\max}$, where $\alpha_{\min} = 1 - \frac{OPT}{c_{j^*} + OPT}$ and $\alpha_{\max} = 1 - \frac{OPT}{m \cdot 2^m (c_{j^*} + OPT)}$.*

Proof. The set that maximizes $f(S) - c(S)$ is also the best response of the agent under contract $\alpha = 1$. We note that since $OPT > 0$, it must be that $\alpha^* < 1$. By Observation 4.1, it holds that

$$(4.10) \quad \frac{OPT}{2^m} \leq (1 - \alpha^*)f(S^*) \leq f(S^*) - c(S^*) \leq OPT,$$

where the second inequality is since the utility of the principal under contract α^* is bounded by the welfare generated by S^* as the utility of the agent is non-negative.

We first show that $\alpha^* \geq \alpha_{\min}$. It holds that

$$\alpha^* \cdot (c(S^*) + OPT) - c(S^*) \geq \alpha^* \cdot (c(S^*) + f(S^*) - c(S^*)) - c(S^*) = \alpha^* \cdot f(S^*) - c(S^*) \geq 0,$$

where the first inequality is by Inequality (4.10), and the last inequality is since the utility of the agent under α^* and S^* is non-negative.

By rearranging, we get that

$$\alpha^* \geq \frac{c(S^*)}{c(S^*) + OPT} \geq \frac{c_{j^*}}{c_{j^*} + OPT} = 1 - \frac{OPT}{c_{j^*} + OPT},$$

as needed.

To show that $\alpha^* \leq \alpha_{\max}$, observe that

$$(1 - \alpha^*)(c(S^*) + OPT) \stackrel{(4.10)}{\geq} (1 - \alpha^*)(c(S^*) + f(S^*) - c(S^*)) = (1 - \alpha^*)f(S^*) \stackrel{(4.10)}{\geq} \frac{OPT}{2^m}.$$

Rearranging this, we obtain

$$\alpha^* \leq 1 - \frac{OPT}{2^m (c(S^*) + OPT)} \leq 1 - \frac{OPT}{m \cdot 2^m (c_{j^*} + OPT)},$$

which concludes the proof. $\square$

We are now ready to prove Theorem 4.2.

Proof. [of Theorem 4.2] Let α^* , S^* be the best contract with its best response. Note that if $u_p(S^*, \alpha^*) \leq 0$, the claim holds trivially because Algorithm 4 ensures that $u_p(S, \alpha) \geq 0$. Otherwise, $u_p(S^*, \alpha^*) = (1 - \alpha^*)f(S^*) > 0$. It must then hold that $\alpha^* < 1$ and $f(S^*) > 0$. So, in particular, $S^* \neq \emptyset$, so that $j^* \in \arg \max_{j \in S^*} c_j$ is well defined. Also note that $OPT = \max_{\hat{S} \subseteq A} (f(\hat{S}) - c(\hat{S})) \geq f(S^*) - c(S^*) \geq (1 - \alpha^*)f(S^*) > 0$, because the agent's utility from S^* is non-negative.

If $c_{j^*} = 0$, then the optimal contract is $\alpha = 0$, which is the contract considered in Step 1 of the algorithm. Otherwise, $c_{j^*} > 0$. Consider the iteration of Algorithm 4 in which $j = j^*$. Let us denote $\alpha_{\min} = 1 - \frac{OPT}{c_{j^*} + OPT}$ and $\alpha_{\max} = 1 - \frac{OPT}{m \cdot 2^m (c_{j^*} + OPT)}$. We claim that then there must be a value of $k \in \{0, \dots, \lceil \log_{1/(1-\epsilon)} m \cdot 2^m \rceil\}$ such that $1 - \alpha_{j,k} \leq 1 - \alpha^* \leq \frac{1 - \alpha_{j,k}}{1 - \epsilon}$. Indeed, for $k = 0$ we have $\frac{1 - \alpha_{j^*,0}}{1 - \epsilon} = 1 - \alpha_{\min} \geq 1 - \alpha^*$, where the inequality follows by Lemma 4.1, while for $k = \lceil \log_{1/(1-\epsilon)} m \cdot 2^m \rceil$ we have $1 - \alpha^* \geq 1 - \alpha_{\max} \geq 1 - \alpha_{j^*, \lceil \log_{1/(1-\epsilon)} m \cdot 2^m \rceil}$, where the first inequality follows again by Lemma 4.1. So there must be a $k \in \{0, \dots, \lceil \log_{1/(1-\epsilon)} m \cdot 2^m \rceil\}$ with the desired properties.

We claim that for this choice of j, k , contract $\alpha_{j,k}$ provides a $(1 - \epsilon)$ -approximation to the optimal contract. To see this, let S be the choice of the agent under $\alpha_{j,k}$. Using Proposition 3.1 in [18] (showing monotonicity of f of the best response as a function of the contract α), since $\alpha_{j,k} \geq \alpha^*$, it must hold that $f(S) \geq f(S^*)$. We thus obtain,

$$(1 - \alpha_{j,k})f(S) \geq (1 - \alpha_{j,k})f(S^*) \geq (1 - \epsilon)(1 - \alpha^*)f(S^*),$$

which completes the proof. $\square$

Algorithm 4 FPTAS for a single agent using value and demand oracles

Parameter: $\epsilon \in (0, 1)$
Input: Costs $c_1, \dots, c_m \in \mathbb{R}_{\geq 0}$, value and demand oracle access to a function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$
Output: A contract α and a best response set S

```

1: Let  $\alpha = 0$  and  $S = \arg \max_{\hat{S}: c(\hat{S})=0} f(\hat{S})$ 
2: Let  $OPT = \max_{\hat{S} \subseteq A} (f(\hat{S}) - c(\hat{S}))$ 
3: for  $j \in A$  with  $c_j > 0$  do
4:   for  $k = 0, \dots, \lceil \log_{1/(1-\epsilon)} m \cdot 2^m \rceil$  do
5:     Let  $\alpha_{j,k} = 1 - (1 - \epsilon)^{k+1} \cdot \frac{OPT}{c_j + OPT}$ 
6:     Let  $S_{j,k} = \arg \max_{\hat{S} \subseteq A} \left( f(\hat{S}) - \sum_{j' \in \hat{S}} \frac{c_{j'}}{\alpha_{j,k}} \right)$ 
7:     if  $(1 - \alpha_{j,k})f(S_{j,k}) > (1 - \alpha)f(S)$  then
8:        $\alpha = \alpha_{j,k}$ 
9:        $S = S_{j,k}$ 
10:    end if
11:  end for
12: end for
13: return  $\alpha, S$ 

```

The query complexity of Algorithm 4 is upper bounded by $O(m \cdot \log_{1/(1-\epsilon)} m \cdot 2^m)$, which is $O(\frac{m^2}{\epsilon})$. We note that by using Proposition 7.1, one can tighten Lemma 4.1 for subadditive rewards, and improve the query complexity to $O(m \cdot \log_{1/(1-\epsilon)} m^2)$, which is $O(\frac{m \log m}{\epsilon})$.

4.2 Making Single-Agent Contracts Robust Algorithm 5 runs, for every agent i' , the FPTAS from Section 4.1, with $\epsilon = 1/2$, which returns a single-agent contract α'_i and best response set S'_i . The algorithm then considers an agent i that maximizes $(1 - \alpha'_i)f(S'_i)$.

A natural extension of this single-agent contract into a multi-agent contract $\vec{\alpha}$ would be to set $\alpha_i = \alpha'_i$ and $\alpha_{i'} = 0$ for all $i' \neq i$. Indeed, this ensures that $S_i = S'_i$ and $S_{i'} = \emptyset$ for all $i' \neq i$ is an equilibrium for $\vec{\alpha}$. However, this contract may admit additional equilibria. In particular, agents $i' \neq i$ may take zero-cost actions, in which case $S_i = S'_i$ may no longer be a best response for agent i .

To address this problem, depending on whether the total value of the zero-cost actions is high or not, the algorithm either returns $\vec{\alpha} = \vec{\epsilon}$ for some small enough $\epsilon > 0$, or further increases α_i to $\alpha_i = \frac{1+\alpha'_i}{2}$ while keeping $\alpha_{i'} = 0$ for all $i' \neq i$. The following lemma shows the robust approximation guarantee that the algorithm obtains this way. The proof once again relies on Lemma 3.1.

LEMMA 4.2. *Let i, α'_i, S'_i be defined as in Step 4 of Algorithm 5, and $\vec{\alpha}, \Lambda$ be the returned contract and guarantee of Algorithm 5. Then, any S that is a best response to $\vec{\alpha}$ guarantees the principal a utility of at least Λ , and it holds that*

$$\Lambda \geq \frac{1}{12}(1 - \alpha'_i)f(S'_i).$$

Proof. We first consider the case where $f(A^0) > \frac{1}{3}(1 - \alpha'_i)f(S'_i)$. Note that A^0 is subset stable under $\vec{\alpha} = \vec{0}$. So, by Lemma 3.1, any equilibrium S with respect to $\vec{\alpha} = \vec{\epsilon}$ satisfies $f(S) \geq \frac{1}{2}f(A^0)$. We thus obtain

$$\left(1 - \sum_{i'} \alpha_{i'}\right) f(S) = (1 - \epsilon)f(S) = \frac{1}{2}f(S) \geq \frac{1}{4}f(A^0) = \Lambda > \frac{1}{12}(1 - \alpha'_i)f(S'_i).$$

Otherwise, $\alpha_i = \frac{1+\alpha'_i}{2}$ and $\alpha_{i'} = 0$ for $i' \neq i$. Note that $\alpha_i > \alpha'_i$ because for the α'_i returned by Algorithm 4 it holds that $\alpha'_i < 1$. Consider any set of actions S that is incentivized by $\vec{\alpha}$. We claim that $f(S) \geq \frac{1}{3}f(S'_i)$. This shows the claim because then

$$\left(1 - \sum_{i'} \alpha_{i'}\right) f(S) = (1 - \alpha_i)f(S) = \frac{1}{2}(1 - \alpha'_i)f(S) \geq \frac{1}{6}(1 - \alpha'_i)f(S'_i) = \Lambda.$$

Algorithm 5 Robust Approximation of Single Agent Benchmark

Input: Costs $c_1, \dots, c_m \in \mathbb{R}_{\geq 0}$, value and demand oracle access to a submodular function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$

Output: A contract $\vec{\alpha}$ with a guarantee of principal's utility of Λ for any equilibrium of $\vec{\alpha}$

```

1: Let  $\vec{\alpha}' = \vec{0}$ 
2: Let  $A^0 = \{j \in A \mid c_j = 0\}$  be the zero cost actions
3: Run Algorithm 4 (with  $\epsilon = \frac{1}{2}$ ) for each agent  $i \in [n]$ , to find  $\alpha'_i$  and best response set  $S'_i$ 
4: Let agent  $i$  be the agent such that  $(1 - \alpha'_i)f(S'_i)$  is highest
5: if  $f(A^0) > \frac{1}{3}(1 - \alpha'_i)f(S'_i)$  then
6:   Let  $\epsilon = \frac{1}{2n}$ 
7:   Let  $\vec{\alpha} = \vec{\epsilon}$ 
8:   Let  $\Lambda = \frac{1}{4}f(A^0)$ 
9: else
10:  Let  $\alpha_i = \frac{1+\alpha'_i}{2}$  and  $\alpha_{i'} = 0$  for all  $i' \neq i$ 
11:  Let  $\Lambda = \frac{1}{6}(1 - \alpha'_i)f(S'_i)$ 
12: end if
13: return  $\vec{\alpha}, \Lambda$ 

```

It remains to show that $f(S) \geq \frac{1}{3}f(S'_i)$. We first use that under $\vec{\alpha}$ agent i weakly prefers S_i over S'_i . This shows that

$$(4.11) \quad \alpha_i f(S) - c(S_i) \geq \alpha_i f(S'_i \cup S_{-i}) - c(S'_i) \geq \alpha_i f(S'_i) - c(S'_i).$$

On the other hand, under $\vec{\alpha}'$ agent i weakly prefers S'_i over S_i and thus

$$\alpha'_i f(S'_i) - c(S'_i) \geq \alpha'_i f(S_i) - c(S_i),$$

which implies

$$(4.12) \quad \alpha'_i f(S'_i) - c(S'_i) \geq \alpha'_i f(S) - \alpha'_i f(S_{-i}) - c(S_i).$$

By summing up (4.11) and (4.12), we get

$$(\alpha_i - \alpha'_i)f(S) \geq (\alpha_i - \alpha'_i)f(S'_i) - \alpha'_i f(S_{-i}),$$

or equivalently

$$f(S) \geq f(S'_i) - \frac{\alpha'_i}{\alpha_i - \alpha'_i} f(S_{-i}).$$

Note that since $\alpha_{i'} = 0$ for all $i' \neq i$, we must have $S_{-i} \subseteq A^0$. We thus have $f(S) \geq f(S'_i) - \frac{\alpha'_i}{\alpha_i - \alpha'_i} f(A^0)$. Moreover, by the definition of α_i , we have $\frac{\alpha'_i}{\alpha_i - \alpha'_i} = \frac{2\alpha'_i}{1 - \alpha'_i} \leq \frac{2}{1 - \alpha'_i}$. Furthermore, $\frac{1}{1 - \alpha'_i} f(A^0) \leq \frac{1}{3}f(S'_i)$. Therefore,

$$f(S) \geq \frac{1}{3}f(S'_i),$$

as claimed. $\square$

We are now ready to prove Theorem 4.1.

Proof. [of Theorem 4.1] The algorithm clearly runs in polynomial time using polynomially many value and demand queries. We claim that Algorithm 5 satisfies the assertion of the theorem. Let $\vec{\alpha}$ be the contract returned by Algorithm 5, and let S be any equilibrium of $\vec{\alpha}$.

Let $\vec{\alpha}^*$ be any contract and any equilibrium S^* of $\vec{\alpha}^*$ such that $\alpha_i^* = 0$ and $S_i^* = \emptyset$ for all but one $i \in [n]$. Then, by Theorem 4.2, α'_i and S'_i as defined in Line 4 of the algorithm, fulfill $(1 - \alpha'_i)f(S'_i) \geq \frac{1}{2}(1 - \alpha_i^*)f(S_i^*)$. So, by Lemma 4.2, we have $\Lambda \geq \frac{1}{24} \cdot (1 - \sum_{i \in [n]} \alpha_i^*)f(S^*)$, as desired. $\square$

5 Putting it all Together

In this section, we wrap up the proof of Main Theorem 1.1. So far, we presented two algorithms: Algorithm 3 gives a constant-factor approximation against the optimal contract in which no agent is large; Algorithm 5 gives an approximation against the best single-agent contract. In this section, we show that restricting to these two cases incurs only a small loss. So, taking the better of the two solutions will give us a constant-factor approximation against the overall optimal contract.

In more detail, the main remaining statement to be shown is the following.

THEOREM 5.1. (REDUCTION TO NO AGENT IS LARGE / SINGLE AGENT) *Consider any submodular success probability function f . Let $\vec{\alpha}^*$ be any contract and S^* be an equilibrium of $\vec{\alpha}^*$. For any $0 \leq \phi \leq 1$, there exists a contract $\vec{\alpha}'$ and an equilibrium S' of $\vec{\alpha}'$ fulfilling $(1 - \sum_i \alpha'_i)f(S') \geq \frac{\phi}{1120} \cdot (1 - \sum_i \alpha_i^*)f(S^*)$ and*

1. *(no large agent) $f(S'_i) \leq \phi f(S')$ for all $i \in [n]$, or*
2. *(single agent) $\alpha'_i = 0$ and $S_i = \emptyset$ for all but one $i \in [n]$.*

We prove Theorem 5.1 below. Before, let's observe that with Theorem 5.1 in hand, the proof of Main Theorem 1.1 becomes straightforward.

Algorithm 6 Robust Approximation for the Optimal Contract

Input: Costs $c_1, \dots, c_m \in \mathbb{R}_{\geq 0}$, value and demand oracle access to a submodular function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$

Output: A contract $\vec{\alpha}$ that approximates the optimal contract

- 1: Run Algorithm 3 on $c_1, \dots, c_m, f$, with parameters $\epsilon = \frac{1}{8n}$, $\rho = \frac{1}{6}$ and receive a contract $\vec{\alpha}_{\text{multi-agent}}$ with guarantee $\Lambda_{\text{multi-agent}}$
 - 2: Run Algorithm 5 on $c_1, \dots, c_m, f$, and receive a contract $\vec{\alpha}_{\text{single-agent}}$ with guarantee $\Lambda_{\text{single-agent}}$
 - 3: **if** $\Lambda_{\text{multi-agent}} > \Lambda_{\text{single-agent}}$ **then**
 - 4: $\vec{\alpha} = \vec{\alpha}_{\text{multi-agent}}$
 - 5: **else**
 - 6: $\vec{\alpha} = \vec{\alpha}_{\text{single-agent}}$
 - 7: **end if**
 - 8: **return** $\vec{\alpha}$
-

Proof. [of Main Theorem 1.1] We show that Algorithm 6 satisfies the assertion of the theorem.

By construction, the returned contract has the property that every equilibrium has principal utility at least $\Lambda = \max\{\Lambda_{\text{multi-agent}}, \Lambda_{\text{single-agent}}\}$. We still have to show that $\Lambda = \Omega(1)u_p(\vec{\alpha}^*, S^*)$.

To this end, let ϕ be the constant from Theorem 3.1. By Theorem 5.1, there exists a contract $\vec{\alpha}'$ with an equilibrium S' fulfilling $(1 - \sum_i \alpha'_i)f(S') \geq \frac{\phi}{1120} \cdot (1 - \sum_i \alpha_i^*)f(S^*)$ and (1) $f(S'_i) \leq \phi f(S')$ for all $i \in [n]$, or (2) $\alpha'_i = 0$ and $S_i = \emptyset$ for all but one $i \in [n]$.

We will distinguish these two cases. In the first case, Theorem 3.1 ensures that $\Lambda_{\text{multi-agent}} \geq \kappa(1 - \sum_i \alpha'_i)f(S') \geq \kappa \frac{\phi}{1120} \cdot (1 - \sum_i \alpha_i^*)f(S^*)$; in the second case, Theorem 4.1 ensures that $\Lambda_{\text{single-agent}} \geq \frac{1}{12}(1 - \sum_i \alpha'_i)f(S') \geq \frac{1}{12} \frac{\phi}{1120} \cdot (1 - \sum_i \alpha_i^*)f(S^*)$. $\square$

Reduction to no agent is large or single agent. The key difficulty in proving Theorem 5.1 is the lack of monotonicity demonstrated in Example 1.1. That is, when removing all agents except for i , this agent might do less. To mitigate this effect, we would like to increase α_i . However, this is problematic if α_i is already close to 1. Therefore, our key lemma for showing Theorem 5.1 establishes that it is sufficient to consider contracts in which all of the α_i are bounded away from 1 or all but one of them are 0.

LEMMA 5.1. *Consider any submodular success probability function f . Let $\vec{\alpha}^*$ be any contract and let S^* be any equilibrium of $\vec{\alpha}^*$. Then there exist a contract $\vec{\alpha}'$ and an equilibrium S' of $\vec{\alpha}'$ fulfilling $(1 - \sum_i \alpha'_i)f(S') \geq \frac{1}{20} \cdot (1 - \sum_i \alpha_i^*)f(S^*)$ and*

1. *$\alpha'_i \leq \frac{3}{4}$ for all $i \in [n]$, or*

2. $\alpha'_i = 0$ and $S_i = \emptyset$ for all but one $i \in [n]$.

Proof. Consider some contract $\bar{\alpha}^*$ and some equilibrium S^* of $\bar{\alpha}^*$. Clearly, we can focus on the case where $\sum_i \alpha_i^* < 1$. Note that in this case there is nothing to show when $\alpha_i^* \leq 3/4$ for all i . Moreover, there can be at most one agent i such that $\alpha_i^* > 3/4$. Let i denote the index of this agent.

Case 1: Let's first consider the case where $(1 - \alpha_i^*)f(S_i^*) \geq 4f(S_{-i}^*)$. Then, because $\alpha_i^* > 3/4$, we have $f(S_i^*) \geq 16f(S_{-i}^*)$. By subadditivity of f , this implies that $f(S_i^*) \geq 16(f(S^*) - f(S_i^*))$, or equivalently, $f(S_i^*) \geq \frac{16}{17}f(S^*)$.

Consider contract $\bar{\alpha}'$ with $\alpha'_i = \frac{1+\alpha_i^*}{2}$ and $\alpha'_{i'} = 0$ for $i' \neq i$. Note that $\alpha'_i > \alpha_i^*$. For agents $i' \neq i$ it's a best response to play $S'_{i'} = \emptyset$ (independent of what agent i does). Let S'_i be any best response by agent i (when $S'_{i'} = \emptyset$ for all $i' \neq i$). We claim that $f(S'_i) \geq \frac{1}{2}f(S_i^*)$. This shows the claim because then

$$\begin{aligned} \left(1 - \sum_{i'} \alpha'_{i'}\right)f(S') &= (1 - \alpha'_i)f(S'_i) = \frac{1}{2}(1 - \alpha_i^*)f(S'_i) \\ &\geq \frac{1}{4}(1 - \alpha_i^*)f(S_i^*) \geq \frac{1}{4}\left(1 - \sum_{i'} \alpha_{i'}^*\right)f(S_i^*) \geq \frac{4}{17}\left(1 - \sum_{i'} \alpha_{i'}^*\right)f(S^*), \end{aligned}$$

where the second equality follows by the definition of α'_i . Note that the sum is over all $i' \in [n]$.

Towards showing that $f(S') \geq \frac{1}{2}f(S_i^*)$, first observe that under $\bar{\alpha}'$ agent i weakly prefers S'_i over S_i^* . Since $S'_{i'} = \emptyset$ for all $i' \neq i$ this shows that

$$(5.13) \quad \alpha'_i f(S'_i) - c(S'_i) \geq \alpha'_i f(S_i^*) - c(S_i^*).$$

On the other hand, under $\bar{\alpha}^*$ agent i weakly prefers S_i^* over S'_i and thus, given the choices S_{-i}^* of the agents other than i , it must hold that

$$\alpha_i^* f(S_i^* | S_{-i}^*) - c(S_i^*) \geq \alpha_i^* f(S'_i | S_{-i}^*) - c(S'_i).$$

By subadditivity of f , we have $f(S_i^* | S_{-i}^*) = f(S_i \cup S_{-i}) - f(S_{-i}) \leq f(S_i^*)$. By monotonicity of f , we have $f(S'_i | S_{-i}^*) = f(S'_i \cup S_{-i}^*) - f(S_{-i}^*) \geq f(S'_i) - f(S_{-i}^*)$. We thus obtain

$$(5.14) \quad \alpha_i^* f(S_i^*) - c(S_i^*) \geq \alpha_i^* f(S'_i) - \alpha_i^* f(S_{-i}^*) - c(S'_i).$$

By summing up (5.13) and (5.14), we get

$$(\alpha'_i - \alpha_i^*)f(S'_i) \geq (\alpha'_i - \alpha_i^*)f(S_i^*) - \alpha_i^* f(S_{-i}^*),$$

or equivalently (using $\alpha'_i > \alpha_i^*$)

$$f(S'_i) \geq f(S_i^*) - \frac{\alpha_i^*}{\alpha'_i - \alpha_i^*} f(S_{-i}^*).$$

By the definition of α'_i , we have $\frac{\alpha_i^*}{\alpha'_i - \alpha_i^*} = \frac{2\alpha_i^*}{1 - \alpha_i^*} \leq \frac{2}{1 - \alpha_i^*}$. Applying the condition of Case 1, we furthermore have $\frac{1}{1 - \alpha_i^*} f(S_{-i}^*) \leq \frac{1}{4}f(S_i^*)$. So, $\frac{\alpha_i^*}{\alpha'_i - \alpha_i^*} f(S_{-i}^*) \leq \frac{1}{2}f(S_i^*)$ and therefore $f(S'_i) \geq \frac{1}{2}f(S_i^*)$. This concludes the argument for this case.

Case 2: Next consider the case where $(1 - \alpha_i^*)f(S_i^*) < 4f(S_{-i}^*)$. In this case we have

$$\begin{aligned} \left(1 - \sum_{i'} \alpha_{i'}^*\right)f(S^*) &= \left(1 - \alpha_i^* - \sum_{i' \neq i} \alpha_{i'}^*\right)(f(S_i^* | S_{-i}^*) + f(S_{-i}^*)) \\ (5.15) \quad &\leq (1 - \alpha_i^*)f(S_i^* | S_{-i}^*) + f(S_{-i}^*) \leq (1 - \alpha_i^*)f(S_i^*) + f(S_{-i}^*) \leq 5f(S_{-i}^*), \end{aligned}$$

where the last inequality follows by the condition of Case 2, and the one before follows by subadditivity of f .

In order to define a contract without agent i , we will use Lemma 3.1. To this end, we will show that the set $S_{-i}^* = S^* \setminus S_i^*$ is subset stable for contract $\bar{\alpha}^*$ (and in fact also for contract $\bar{\beta}$ which is equal to $\bar{\alpha}^*$ in all coordinates but coordinate i , where $\beta_i = 0$).

To see this, observe that for all $i' \neq i$ and all $S'_{i'} \subseteq S^*_{i'}$ we have

$$\alpha^*_{i'} f(S^*) - c(S^*) \geq \alpha^*_{i'} f(S'_{i'}, S^*_{-i'}) - c(S'_{i'}),$$

or equivalently

$$\alpha^*_{i'} f(S^*_{i'} \setminus S'_{i'} \mid S'_{i'} \cup S^*_{-i'}) \geq c(S^*_{i'}) - c(S'_{i'}).$$

By submodularity of f , we have $f(S^*_{i'} \setminus S'_{i'} \mid S'_{i'} \cup S^*_{-i'}) \leq f(S^*_{i'} \setminus S'_{i'} \mid S'_{i'} \cup (S^*_{-i'} \setminus S^*_i))$. Moreover, since $S'_{i'} \subseteq S^*_{i'}$, it holds that $f(S^*_{i'} \setminus S'_{i'} \mid S'_{i'} \cup (S^*_{-i'} \setminus S^*_i)) = f(S^*_{i'} \cup (S^*_{-i'} \setminus S^*_i)) - f(S'_{i'} \cup (S^*_{-i'} \setminus S^*_i))$. We thus obtain

$$\alpha^*_{i'} f(S^*_{i'}, S^*_{-i'} \setminus S^*_i) - c(S^*) \geq \alpha^*_{i'} f(S'_{i'}, S^*_{-i'} \setminus S^*_i) - c(S'_{i'}).$$

Consider contract $\vec{\beta}$ with $\beta_{i'} = \alpha^*_{i'}$ for $i' \neq i$ and $\beta_i = 0$. Since $\alpha^*_i > 3/4$, we can choose $\epsilon > 0$ small enough so that $\sum_{i' \neq i} 2\alpha^*_{i'} + n\epsilon \leq 1/2$. Consider any equilibrium $S^\dagger$ with respect to $\vec{\alpha}' = 2\vec{\beta} + \vec{\epsilon}$. Then, by Lemma 3.1, we have $f(S^\dagger) \geq \frac{1}{2}f(S^*_{-i})$. So, the principal utility of contract $2\vec{\beta} + \vec{\epsilon}$ in equilibrium $S^\dagger$ is

$$\left(1 - \sum_{i' \neq i} 2\alpha^*_{i'} - n\epsilon\right) f(S^\dagger) \geq \frac{1}{4}f(S^*_{-i}) \geq \frac{1}{20} \left(1 - \sum_{i'} \alpha^*_{i'}\right) f(S^*),$$

where for the first inequality we used that $f(S^\dagger) \geq \frac{1}{2}f(S^*_{-i})$ and $\sum_{i' \neq i} 2\alpha^*_{i'} + n\epsilon \leq 1/2$. The second inequality holds by Inequality (5.15). As we have $\sum_{i'} \alpha'_{i'} = \sum_{i' \neq i} 2\alpha^*_{i'} + n\epsilon \leq 1/2$, in particular, we also have $\alpha'_{i'} \leq 3/4$ for all i' . This completes the argument for this case. $\square$

We are now ready to prove Theorem 5.1.

Proof. [of Theorem 5.1] By Lemma 5.1, there is a contract α' incentivizing S' such that $(1 - \sum_i \alpha'_i)f(S') \geq \frac{1}{20} \cdot (1 - \sum_i \alpha^*_i)f(S^*)$ and $\alpha'_i \leq \frac{3}{4}$ for all $i \in [n]$, or $\alpha'_i = 0$ and $S'_i = \emptyset$ for all but one $i \in [n]$.

In the latter case, there is nothing to be shown. We are done as well if $f(S'_i) \leq \phi f(S')$ for all $i \in [n]$. Otherwise, there is an agent i such that $f(S'_i) > \phi f(S')$ and $\alpha'_i \leq \frac{3}{4}$. Now consider the contract $\vec{\alpha}''$ that sets $\alpha''_i = \frac{7}{8}$ for this agent i and $\alpha''_{i'} = 0$ for all agents $i' \neq i$. For agents $i' \neq i$ it is a best response to play $S''_{i'} = \emptyset$. Let S''_i be any best response by agent i under this choice of $S''_{i'}$ for $i' \neq i$.

We now have

$$\frac{7}{8}f(S''_i) \geq \alpha''_i f(S''_i) - c(S''_i) \geq \alpha''_i f(S'_i) - c(S'_i) \geq \frac{1}{8}f(S'_i) + \alpha'_i f(S'_i) - c(S'_i) \geq \frac{1}{8}f(S'_i).$$

So,

$$(1 - \alpha''_i)f(S''_i) = \frac{1}{8}f(S''_i) \geq \frac{1}{56}f(S'_i) \geq \frac{\phi}{56}f(S') \geq \frac{\phi}{56}(1 - \sum_{i'} \alpha'_{i'})f(S') \geq \frac{\phi}{1120}(1 - \sum_{i'} \alpha^*_{i'})f(S^*),$$

as claimed. $\square$

6 No PTAS for Multiple Agents, Not Even with Demand Queries

In this section, we show that even in the case of binary actions (the model studied in [19]), there is an absolute constant $\eta < 1$ for which no algorithm can guarantee an approximation of η with a polynomial number of value and demand queries even under the best equilibrium. This result strengthens a result by [27], who showed that no poly-time algorithm can get 0.7 approximation with access to value oracle alone.

THEOREM 6.1. *There exists a constant $\eta < 1$ for which any algorithm that uses a sub-exponential number of value and demand queries to the submodular success probability function returns a η -approximate optimal contract with an exponentially small probability in n .*

Our proof relies on the construction of a family of functions $\{f_T : 2^A \rightarrow R_{\geq 0}\}_T$, defined using the following function, parameterized by k :

$$f(x, y) = \begin{cases} \sqrt{k} & \text{if } x + y \geq k \\ \sqrt{x + y} & \text{if } x + y < k \text{ and } y \leq \frac{k}{2} \\ \sqrt{x + \frac{k}{2}} + \frac{y - \frac{k}{2}}{\sqrt{k} + \sqrt{x + \frac{k}{2}}} & \text{otherwise,} \end{cases}$$

where

$$f_T(S) = f(|S \setminus T|, |S \cap T|) \text{ with parameter } k = |T|.$$

The construction of the hard example is different in nature than the constructions given in [19, 27] for XOS functions. The main idea in [19] is to hide a single good set of agents with a higher value which cannot be learned efficiently using value and demand queries. This increased value decreases the payments to all agents, leading to an attractive set of agents. However, in order to make all its subsets non-attractive, the construction heavily uses the non-monotonicity of the marginals in XOS functions. Therefore, this approach cannot be extended to submodular functions. The hardness result of [27] for XOS functions in the case of value queries also relies on hiding a small set of good agents with high value, which can be incentivized by relatively low payments. However, since the attractive set has a much higher value than a random set of the same size, this type of hidden set can be learned relatively easily by demand queries. Therefore, establishing a hardness result in our case requires new ideas.

Our construction is based on creating a good set of agents whose value is indistinguishable from other sets of the same size (so demand queries would not be useful in identifying the hidden set). Thus, the only way to make the set more attractive is by making it cheaper to incentivize. This is done by decreasing the value of its subsets (instead of increasing the value of the set itself as was done in the former constructions). This decrease in value makes the marginal contribution of every agent higher, which in turn decreases the payments necessary to incentivize them, while keeping the attractive set (and all its subsets) hidden, as demand oracles are better in finding sets with higher values than sets with lower values.

CLAIM 6.1. *For every set $T \subset A$ of even size, the function f_T is monotone and submodular.*

Proof. Let

$$f_x(x, y) = f(x + 1, y) - f(x, y) = \begin{cases} 0 & \text{if } x + y \geq k \\ \frac{1}{\sqrt{x+y+1} + \sqrt{x+y}} & \text{if } x + y < k, y \leq \frac{k}{2} \\ \frac{1}{\sqrt{x+1+\frac{k}{2}} + \sqrt{x+\frac{k}{2}}} + \frac{y - \frac{k}{2}}{\sqrt{k} + \sqrt{x+1+\frac{k}{2}}} - \frac{y - \frac{k}{2}}{\sqrt{k} + \sqrt{x+\frac{k}{2}}} & \text{otherwise} \end{cases}$$

and

$$f_y(x, y) = f(x, y + 1) - f(x, y) = \begin{cases} 0 & \text{if } x + y \geq k \\ \frac{1}{\sqrt{x+y+1} + \sqrt{x+y}} & \text{if } x + y < k, y < \frac{k}{2} \\ \frac{1}{\sqrt{k} + \sqrt{x+\frac{k}{2}}} & \text{otherwise.} \end{cases}$$

To show that f_T is monotone, it is equivalent to prove that $f_x(x, y), f_y(x, y) \geq 0$ for every x, y . The only case that is not trivial is for f_x , where we are in the third case for which

$$\begin{aligned} f_x(x, y) &= \frac{1}{\sqrt{x+1+\frac{k}{2}} + \sqrt{x+\frac{k}{2}}} + \underbrace{\frac{y - \frac{k}{2}}{\sqrt{k} + \sqrt{x+1+\frac{k}{2}}} - \frac{y - \frac{k}{2}}{\sqrt{k} + \sqrt{x+\frac{k}{2}}}}_{\leq 0} \\ &\geq \frac{1}{\sqrt{x+1+\frac{k}{2}} + \sqrt{x+\frac{k}{2}}} + \frac{\frac{k}{2} - x}{\sqrt{k} + \sqrt{x+1+\frac{k}{2}}} - \frac{\frac{k}{2} - x}{\sqrt{k} + \sqrt{x+\frac{k}{2}}} \\ &= \frac{1}{\sqrt{k} + \sqrt{1+\frac{k}{2}+x}} \geq 0, \end{aligned}$$

where the first inequality holds since in this case, $x + y \leq k$.

To show submodularity, it is sufficient to show that the derivative of all cases of f_x and f_y with respect to both x, y are non-positive, and that f_x and f_y decrease in the transitions between the cases of the function. The only non-trivial case for the derivative is again the third case of f_x for which

$$\begin{aligned} \frac{f_x(x, y)}{\delta x} &= \underbrace{\frac{-1}{\sqrt{k/2 + x}\sqrt{2 + k + 2x}(\sqrt{k + 2x} + \sqrt{2 + k + 2x})}}_{\text{increasing in } x} \\ &+ \underbrace{\frac{y - \frac{k}{2}}{2\sqrt{\frac{k}{2} + x}(\sqrt{k} + \sqrt{\frac{k}{2} + x})^2} - \frac{y - \frac{k}{2}}{2\sqrt{1 + \frac{k}{2} + x}(\sqrt{k} + \sqrt{1 + \frac{k}{2} + x})^2}}_{\text{decreasing in } x \text{ and increasing in } y} \\ &\leq -\frac{1}{\sqrt{k}\sqrt{2k}(\sqrt{2k} + \sqrt{2k})} + \frac{\frac{k}{2}}{2\sqrt{\frac{k}{2}}(\sqrt{k} + \sqrt{\frac{k}{2}})^2} - \frac{\frac{k}{2}}{2\sqrt{1 + \frac{k}{2}}(\sqrt{k} + \sqrt{1 + \frac{k}{2}})^2} \leq 0, \end{aligned}$$

where for the first inequality we use that $x \in [0, \frac{k}{2}]$ and that $y \leq k$, and the last inequality holds for every $k > 0$.

We also need to show that in the transitions between the cases, the submodularity is maintained. That is, we need to show that (1) $f_x(x, k/2) \geq f_x(x, k/2 + 1)$ for all $x \leq k/2$, and (2) that $f_y(x, k/2 - 1) \geq f_y(x, k/2)$ for all $x \leq k/2$. The first inequality holds since

$$\begin{aligned} f_x(x, k/2) &= \frac{1}{\sqrt{x + 1 + \frac{k}{2}} + \sqrt{x + \frac{k}{2}}} \\ &\geq \underbrace{\frac{1}{\sqrt{x + 1 + \frac{k}{2}} + \sqrt{x + \frac{k}{2}}} + \frac{1}{\sqrt{k} + \sqrt{x + 1 + \frac{k}{2}}} - \frac{1}{\sqrt{k} + \sqrt{x + \frac{k}{2}}}}_{\leq 0} = f_x(x, k/2 + 1), \end{aligned}$$

and the second inequality holds since

$$f_y(x, k/2 - 1) = \frac{1}{\sqrt{x + \frac{k}{2}} + \sqrt{x + \frac{k}{2} - 1}} \geq \frac{1}{\sqrt{k} + \sqrt{x + \frac{k}{2}}} = f_y(x, k/2),$$

where the inequality is since $x \leq \frac{k}{2}$. $\square$

Consider an instance defined by f_T for some set T where $c_i = \frac{1}{8k^{3/2}}$ for every $i \in A$. Let g_T be the corresponding principal's utility function with respect to f_T .

LEMMA 6.1. *For any set X with $|X \cap T| \leq \frac{k}{2}$, we have $g_T(X) < (0.97 + \frac{1}{3\sqrt{k}}) \cdot \max_{X^*} g_T(X^*)$.*

Proof. We first show that $\max_{X^*} g_T(X^*) > 0.78\sqrt{k}$. In particular, we show that $g_T(T) > 0.78\sqrt{k}$. First observe that $f_T(T) = \sqrt{k}$. Furthermore, for every $i \in T$, we have

$$f_T(T \setminus \{i\}) = \sqrt{\frac{k}{2}} + \left(\sqrt{k} - \sqrt{\frac{k}{2}} \right) \frac{k - 1 - \frac{k}{2}}{\frac{k}{2}} = \sqrt{\frac{k}{2}} + \sqrt{k} \left(1 - \frac{1}{\sqrt{2}} \right) \left(1 - \frac{2}{k} \right).$$

So the marginal of every element is given by

$$f_T(i \mid T \setminus \{i\}) = f_T(T) - f_T(T \setminus \{i\}) = \left(1 - \frac{1}{\sqrt{2}} \right) \frac{2}{\sqrt{k}} = \frac{2 - \sqrt{2}}{\sqrt{k}},$$

implying

$$\frac{c_i}{f_T(i \mid T \setminus \{i\})} = \frac{1}{8k^{3/2}} \frac{\sqrt{k}}{2 - \sqrt{2}} = \frac{1}{k} \frac{1}{8(2 - \sqrt{2})}$$

and so

$$1 - \sum_{i \in T} \frac{c_i}{f_T(i \mid T \setminus \{i\})} = 1 - \frac{1}{8(2 - \sqrt{2})} > 0.78.$$

This implies that

$$g_T(T) = \left(1 - \sum_{i \in T} \frac{c_i}{f_T(i \mid T \setminus \{i\})}\right) f_T(T) > 0.78\sqrt{k}.$$

Given that $g_T(T) > 0.78\sqrt{k}$, to prove the lemma it suffices to show that $g_T(X) \leq \frac{3}{4}\sqrt{k} + \frac{1}{4}$.

If $|X \cap T| \leq \frac{k}{2}$ and $|X| \leq k$, we have for all $i \in X$

$$\frac{c_i}{f_T(i \mid X \setminus \{i\})} = \frac{c_i}{\sqrt{|X|} - \sqrt{|X| - 1}} = c_i(\sqrt{|X|} + \sqrt{|X| - 1}) \geq 2c_i\sqrt{|X| - 1} \geq 2c_i(\sqrt{|X|} - 1).$$

Therefore,

$$\begin{aligned} g_T(X) &= \left(1 - \sum_{i \in X} \frac{c_i}{f_T(i \mid X \setminus \{i\})}\right) f_T(X) \leq (1 - |X|2c_i(\sqrt{|X|} - 1))\sqrt{|X|} \\ &= \sqrt{|X|} - \frac{|X|^2}{4k^{3/2}} + \frac{|X|^{3/2}}{4k^{3/2}} \leq \frac{3}{4}\sqrt{k} + \frac{1}{4}, \end{aligned}$$

as desired, where we use that $x \mapsto \sqrt{x} - \frac{x^2}{4k^{3/2}}$ is non-decreasing for $x \in [0, k]$.

Furthermore, if $|X| > k$, then $f_T(i \mid X \setminus \{i\}) = 0$ and so $g_T(X) = -\infty$. $\square$

We next show how to implement a demand query for f_T using access to value queries for f_T .

LEMMA 6.2. *For every set $T \subseteq A$ of even size $4 \leq k < |A|$, answering a demand query for f_T can be done with polynomially many value queries to f_T (without knowing the set T).*

Proof. We present an algorithm that answers a demand set using poly many value queries.

Let $p_1, \dots, p_n$ be the price vector whose demand is desired; that is, we want to find a set S that maximizes $f_T(S) - \sum_{j \in S} p_j$. Without loss of generality assume that $p_1 \leq p_2 \leq \dots \leq p_n$.

We first observe that the algorithm can learn k (i.e., the size of T), by the single value query $f_T(A)$. Indeed, since $T \subseteq A$, $f_T(A) = \sqrt{k}$.

Next, the algorithm issues value queries of the form $f_T([i])$ for all $i \in [k]$. If $f_T([i]) = \sqrt{i}$ for all $i \in [k]$, then the algorithm returns the set S that maximizes $f_T(S) - \sum_{j \in S} p_j$ among all these sets. This must be a demand set since $[i]$ is the cheapest set of size i , no set of size i has value greater than $\sqrt{i}$, and there is always a demand set of size at most k .

Otherwise, there exists $i \in [k]$ such that $f_T([i]) \neq \sqrt{i}$. We show that in this case the algorithm can find the set T with poly-many additional value queries. We then show that if the algorithm knows T then it can find a demand set with poly-many value queries.

Let i^* be the minimal i such that $f_T([i]) \neq \sqrt{i}$. By the definition of f_T , we get that $i^* < k$ and $|[i^*] \cap T| > \frac{k}{2}$. Since $[i^* - 1] = \sqrt{i^* - 1}$ and that $i^* - 1 < k$, it must be that $|[i^* - 1] \cap T| \leq \frac{k}{2}$, which implies that $i^* \in T$.

Then, for each $j \notin [i^*]$ the algorithm issues the query $f_T([i^* - 1] \cup \{j\})$. If for j it holds that $f_T([i^* - 1] \cup \{j\}) = f_T([i^*])$, it means that either $j \in T$, otherwise, $f_T([i^* - 1] \cup \{j\}) = \sqrt{i^*}$ and $j \notin T$. Then, let j^* be an arbitrary action in $i \setminus [i^*]$ (such exists since $i^* < k$). Then for each $j \in [i^*]$ the algorithm issues the query $f_T([i^*] \cup \{j^*\} \setminus \{j\})$, for which if $f_T([i^*] \cup \{j^*\} \setminus \{j\}) = f_T([i^*])$ then $j \in T$ iff $j^* \in T$. Since whether j^* in T is already known, it means that we know for each $i \in A$ whether it is in T or not.

Once the identity of T is known, one can answer a demand query as follows. Go over all options for x, y satisfying $0 \leq x + y \leq k$, and consider the set S composed of the x cheapest elements in $A \setminus T$, and the y cheapest elements in T . The set S among these sets that maximizes $f_T(S) - \sum_{j \in S} p_j$ is a demand.

This algorithm returns a demand set in polynomial time using $O(|A|)$ value queries, completing the proof of the lemma. $\square$

We are now ready to present the proof of Theorem 6.1.

Proof. [of Theorem 6.1] To prove the theorem, it suffices to show that the statement holds with respect to a deterministic algorithm against a randomized input, where the set T that defines the function f_T is chosen uniformly at random from all sets of size $k = 2 \cdot \lfloor \frac{n}{10} \rfloor$. (Note that restricting the sets T that the adversary chooses from only strengthens our statement.)

By Lemma 6.2, it suffices to consider a deterministic algorithm that issues only value queries, and we may assume without loss of generality that the returned set is queried (otherwise, just add this query at the end of the algorithm).

By Lemma 6.1, for large enough k , in order to get better than $\eta = 0.98$ approximation, for large enough n , the algorithm must ask a value query for a set S for which $|S| \leq k$ and $|S \cap T| > \frac{k}{2}$. We call such a query a *good* query.

Since the probability of a query S to be good is increasing in the size of the set S (up to size k), the probability that a single query is successful is upper bounded by the probability that a hypergeometric random variable with parameters (n, k, k) is greater than $k/2$. For our choice of k (namely, $k = 2 \cdot \lfloor \frac{n}{10} \rfloor$), this probability is exponentially small in n .

We note that, since we bound the probability that the algorithm issues a good query, it is without loss of generality to assume that the queries are non-adaptive. Indeed, at any point in time, it can be assumed that none of the previous queries were good, and for queries that are not good, no information beyond this fact is learned.

In summary, every query is good with exponentially small probability, thus, by the union bound, the probability of issuing a good query is exponentially small for a sub-exponential number of queries. This concludes the proof. $\square$

We remark that our proof applies to submodular success probability functions, but does not rule out the existence of a PTAS (or FPTAS) for the special case of gross substitutes functions. Indeed, the following observation shows that our construction does not satisfy the gross substitutes condition.

OBSERVATION 6.1. *For every $T \subset A$ of even size $k \geq 4$, the function f_T is not gross substitutes.*

Proof. According to the triplet condition in [43], a function $g : 2^A \rightarrow \mathbb{R}_{\geq 0}$ is gross substitutes if and only if it is submodular and for any set S , and any three elements $\{i, j, k\} \notin S$, it holds that

$$\max\{g(i, k | S) + g(j | S), g(j, k | S) + g(i | S)\} \geq g(i, j | S) + g(k | S).$$

Consider two arbitrary elements $i_1, i_2 \in T$, and arbitrary element $j \notin T$, and let $S \subseteq T \setminus \{i_1, i_2\}$ be an arbitrary set of size $k/2$.

The triplet condition requires that

$$\max\{f_T(i_1, j | S) + f_T(i_2 | S), f_T(i_2, j | S) + f_T(i_1 | S)\} \geq f_T(i_1, i_2 | S) + f_T(j | S),$$

which is equivalent to

$$f(1, \frac{k}{2} + 1) + f(0, \frac{k}{2} + 1) \geq f(1, \frac{k}{2}) + f(0, \frac{k}{2} + 2).$$

However,

$$\begin{aligned} f(1, \frac{k}{2} + 1) + f(0, \frac{k}{2} + 1) &= \sqrt{\frac{k}{2} + 1} + \frac{1}{\sqrt{k} + \sqrt{\frac{k}{2} + 1}} + \sqrt{\frac{k}{2}} + \frac{1}{\sqrt{k} + \sqrt{\frac{k}{2}}} \\ &< \sqrt{\frac{k}{2} + 1} + \sqrt{\frac{k}{2}} + \frac{2}{\sqrt{k} + \sqrt{\frac{k}{2}}} \\ &= f(1, \frac{k}{2}) + f(0, \frac{k}{2} + 2), \end{aligned}$$

which concludes the proof. $\square$

7 Guarantees Against the First Best

In this section, we analyze the performance of the best contract against the first best benchmark of social welfare when ignoring incentive constraints, i.e., $OPT = \max_S (f(S) - c(S))$.

We first show that when there is only one agent with m actions and a subadditive reward function, then the principal's utility of the optimal contract is at least $\frac{OPT}{m}$.

PROPOSITION 7.1. *For every subadditive reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ over $|A| = m$ actions and a single agent with costs $c_1, \dots, c_m$, there exists a contract α that guarantees a utility for the principal of at least $\frac{OPT}{m}$, where $OPT = \max_S (f(S) - c(S))$.*

Proof. For this proof, we consider the definition of critical points defined in [18]. A critical point is a value of $\alpha \in (0, 1)$ for which the best responses to contract $\alpha + \epsilon$ are different than the best responses to contract $\alpha - \epsilon$ for every $\epsilon > 0$. In [18], it was shown that this set is always finite. Furthermore, if we order these critical points and denote them by $(0 = \alpha_0 < \alpha_1, \dots, \alpha_k \leq 1)$, then there exists a sequence of action sets $S_0, S_1, \dots, S_k$ such that for each $i = 1, \dots, k$, the sets S_i, S_{i-1} are best responses to contract α_i , the set S_0 is the best response contract $\alpha_0 = 0$ that maximizes the principal utility (i.e., $S_0 = \{j \in A \mid c_j = 0\}$). Moreover, the set S_k is the best response under contract $\alpha = 1$, it holds that $f(S_0) < f(S_1) < \dots < f(S_k)$, and for every $i = 1, \dots, k$ it holds that

$$(7.16) \quad \alpha_i = \frac{c(S_i) - c(S_{i-1})}{f(S_i) - f(S_{i-1})}.$$

Based on these observations, defining $\alpha_{k+1} = 1$, the social welfare can be rewritten by

$$\begin{aligned} OPT &= f(S_k) - c(S_k) = f(S_0) + \sum_{i=1}^k f(S_i) - f(S_{i-1}) - (c(S_i) - c(S_{i-1})) \\ &\stackrel{(7.16)}{=} f(S_0) + \sum_{i=1}^k f(S_i) - f(S_{i-1}) - \alpha_i (f(S_i) - f(S_{i-1})) \\ &= f(S_0) + \sum_{i=1}^k (1 - \alpha_i) (f(S_i) - f(S_{i-1})) \\ &= f(S_0) + \sum_{i=1}^k \left(\sum_{t=i}^k \alpha_{t+1} - \alpha_t \right) (f(S_i) - f(S_{i-1})) \\ &= f(S_0) + \sum_{t=1}^k \sum_{i=1}^t (\alpha_{t+1} - \alpha_t) (f(S_i) - f(S_{i-1})) \\ &= f(S_0) + \sum_{t=1}^k (\alpha_{t+1} - \alpha_t) (f(S_t) - f(S_0)) = \sum_{t=0}^k (\alpha_{t+1} - \alpha_t) f(S_t), \end{aligned}$$

where the first equality is since S_k is the best response under contract $\alpha = 1$, which by definition maximizes the social welfare; the second equality is by telescoping and since $c(S_0) = 0$.

Let $\ell_1 = 0$ if $S_0 \neq \emptyset$, otherwise let $\ell_1 = 1$. So S_{ℓ_1} is the first non-empty set. For $i > 1$ let $\ell_i = \min\{\ell > \ell_{i-1} \mid S_\ell \not\subseteq \cup_{j=1}^{\ell_{i-1}} S_j\}$ (if no such ℓ exists, then let $\ell_i = \infty$). That is, ℓ_i is the smallest $\ell \geq \ell_{i-1}$ such that S_ℓ contains an action that is not contained in any $S_1, \dots, S_{\ell_{i-1}}$. Let d be the last index for which $\ell_d \neq \infty$, and let $T_i = \cup_{j=1}^{\ell_i} S_j$. Note that $d \leq m$. For convenience, let $\alpha_{\ell_{d+1}} = 1$, and let $T_0 = \emptyset$. Then, we can bound the

expression for social welfare derived above by

$$\begin{aligned}
 OPT &= \sum_{t=0}^k (\alpha_{t+1} - \alpha_t) f(S_t) \\
 &= \sum_{i=1}^d \sum_{t=\ell_i}^{\ell_{i+1}-1} (\alpha_{t+1} - \alpha_t) f(S_t) \leq \sum_{i=1}^d (\alpha_{\ell_{i+1}} - \alpha_{\ell_i}) f(T_i) \\
 &\leq \sum_{i=1}^d (\alpha_{\ell_{i+1}} - \alpha_{\ell_i}) \sum_{j=1}^i f(T_j \setminus T_{j-1}) = \sum_{j=1}^d f(T_j \setminus T_{j-1}) \sum_{i=j}^d (\alpha_{\ell_{i+1}} - \alpha_{\ell_i}) \\
 (7.17) \quad &= \sum_{j=1}^d f(T_j \setminus T_{j-1}) (1 - \alpha_{\ell_j}),
 \end{aligned}$$

where the first inequality is since for every $t < \ell_{i+1}$, $S_t \subseteq T_i$; and the second inequality is by subadditivity.

On the other hand, since $\cup_{j=1}^{\ell_i} S_j = T_i$, and $\cup_{j=1}^{\ell_{i-1}} S_j = T_{i-1}$ we get that $T_i \setminus T_{i-1} \subseteq S_{\ell_i}$, which by combining with monotonicity of f it implies that $f(S_{\ell_i}) \geq f(T_i \setminus T_{i-1})$.

So, in total, the best contract among $\alpha_{\ell_1}, \alpha_{\ell_d}$ gives a principal utility of

$$\max_{i \in [d]} (1 - \alpha_{\ell_i}) f(S_{\ell_i}) \geq \frac{1}{d} \sum_{i=1}^d (1 - \alpha_{\ell_i}) f(T_i \setminus T_{i-1}) \geq \frac{1}{d} OPT.$$

Since $d \leq m$, this shows the claim. $\square$

Notably, Proposition 7.1 is tight. Indeed, it was shown in [25] that even in the non-combinatorial model (which corresponds to f being of the form $f(S) = \max_{j \in S} f(j)$), the gap between the social welfare and the maximum principal's utility can be at least m .

We next show how to generalize Proposition 7.1 to the case of multiple agents.

THEOREM 7.1. *For every subadditive reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ over $|A| = m$ actions and multiple agents with action costs $c_1, \dots, c_m$, there exists a contract $\vec{\alpha}$ with a corresponding equilibrium in which the utility for the principal is at least $\frac{OPT}{m}$.*

Proof. For every agent i , let $m_i = |A_i|$. Let S^* be the set that maximizes the social welfare. By Proposition 7.1 we know that for every agent i exists contract α_i that guarantees for the principal a utility of at least $\frac{f(S_i^*) - c(S_i^*)}{m_i}$ assuming that all the other agents do nothing. Consider a distribution over contracts, which chooses agent i with probability $\frac{m_i}{m}$, offering him a contract α_i and all other agents receive 0. Under the best equilibrium of each such contract, we get that the expected utility of the principal is at least

$$(7.18) \quad \sum_i \frac{m_i}{m} \frac{f(S_i^*) - c(S_i^*)}{m_i} = \sum_i \frac{f(S_i^*) - c(S_i^*)}{m} \geq \frac{f(S^*) - c(S^*)}{m} = \frac{OPT}{m}.$$

Thus, there exists i^* such that the best contract (where only agent i^* takes actions and receive payments) achieves a principal's utility of at least $\frac{OPT}{m}$, which concludes the proof. $\square$

Recall that if f is submodular our algorithm ensures (as stated in Theorem 5.1) that *every* equilibrium of the computed contract is a constant-factor approximation of the *best* equilibrium of the *best* contract. As an immediate corollary of Theorem 7.1, we therefore obtain the following bound that also lower-bounds the worst equilibrium.

COROLLARY 7.1. *For every submodular reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ over $|A| = m$ actions and multiple agents with action costs $c_1, \dots, c_m$, there exists a contract $\vec{\alpha}$ for which every corresponding equilibrium guarantees a utility for the principal of at least $\Omega(\frac{OPT}{m})$.*

For completeness, we next show that for the multiple agent case, the gap between the social welfare and the optimal principal's utility can be as large as $\Omega(n)$ even in the binary-action model of [19] and even under additive reward functions. It was shown in [8] that this gap is generally unbounded when the reward function is supermodular.

EXAMPLE 7.1. Consider a setting with n agents with binary actions where the reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ is $f(S) = \frac{|S|}{n}$ while the cost for every agent is $\frac{1}{2n}$. The social welfare of this instance is when all agents exert effort leading to a welfare of $\frac{n}{n} - n \cdot \frac{1}{2n} = \frac{1}{2}$. However, no contract can guarantee the principal a utility of more than $\frac{1}{2n}$ as the principal needs to pay each agent at least $\alpha_i \geq \frac{1}{2}$ for him to exert effort, thus the principal cannot incentivize more than one agent while having a positive utility. The optimal contract that incentivizes at most one agent achieves a principal's utility of $\frac{1}{2n}$.

If we assume f only be subadditive but not necessarily submodular, the gap to the welfare under worst-case equilibrium can indeed be larger. We first give a positive result, showing that the gap is no larger than $O(n \cdot m)$.

PROPOSITION 7.2. For every subadditive reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ over $|A| = m$ actions and n agents with action costs $c_1, \dots, c_m$, there exists a contract α for which every corresponding equilibrium guarantees a utility for the principal of at least $\Omega(\frac{OPT}{n \cdot m})$.

Proof. Let A^0 be the set of actions with a cost of 0, let $A_i^0 = A_i \cap A^0$, let $m_i = |A_i|$, let $i_0^* = \arg \max_{i \in [n]} f(A_i^0)$, let $OPT_i = \max\{f(S_i) - c(S_i) \mid S_i \subseteq A_i\}$, and let $i^* = \arg \max_{i \in [n]} \frac{OPT_i}{m_i}$. We know by Equation (7.18) that $OPT_{i^*} \geq \frac{OPT}{m}$. We consider two cases: If $f(A_{i_0^*}^0) \geq \frac{OPT}{4n \cdot m}$, then we analyze the equilibria of contract $\vec{\alpha}$ where agent i_0^* is offered $\epsilon > 0$, and all other agents are offered 0. Let S be some equilibrium with respect to $\vec{\alpha}$. Assume towards contradiction that $f(S) < f(A_{i_0^*}^0)$. Then agent i_0^* can deviate to $A_{i_0^*}^0$ which will strictly increase his utility since

$$\epsilon \cdot f(S_{-i_0^*} \cup A_{i_0^*}^0) - c(A_{i_0^*}^0) \geq \epsilon \cdot f(A_{i_0^*}^0) > \epsilon \cdot f(S) \geq \epsilon \cdot f(S) - c(S_{i_0^*}).$$

So in this case, the principal's utility under equilibrium S , is at least $(1 - \epsilon)f(A_{i_0^*}^0) = \Omega(\frac{OPT}{n \cdot m})$.

Now consider the case that $f(A_{i_0^*}^0) < \frac{OPT}{4n \cdot m}$. We know by Proposition 7.1 that if assuming that all agents but i^* do nothing, then there exists a contract $\alpha_{i^*}^*$ with a corresponding best response $S_{i^*}^*$ that gives the principal a utility of at least $\frac{OPT_{i^*}}{m_{i^*}}$. Consider a contract $\vec{\alpha}$ where $\alpha_{i^*} = \frac{1 + \alpha_{i^*}^*}{2}$, and $\alpha_i = 0$ for $i \neq i^*$. Let S be an equilibrium with respect to $\vec{\alpha}$. Assume towards contradiction that $f(S) < \frac{f(S_{i^*}^*)}{2}$ then we know that

$$(7.19) \quad \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \geq \alpha_{i^*}^* f(S_{i^*}) - c(S_{i^*}).$$

On the other hand, the utility of agent i^* satisfies

$$\begin{aligned} \alpha_{i^*} f(S) - c(S_{i^*}) &\leq \alpha_{i^*} (f(S_{i^*}) + f(S_{-i^*})) - c(S_{i^*}) \\ &\leq f(S_{-i^*}) + \alpha_{i^*} f(S_{i^*}) - c(S_{i^*}) \\ &= f(S_{-i^*}) + \frac{1 - \alpha_{i^*}^*}{2} f(S_{i^*}) + \alpha_{i^*}^* f(S_{i^*}) - c(S_{i^*}) \\ &\stackrel{(7.19)}{\leq} f(S_{-i^*}) + \frac{1 - \alpha_{i^*}^*}{2} f(S_{i^*}) + \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \\ &\leq f(A^0) + \frac{1 - \alpha_{i^*}^*}{2} f(S) + \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \\ &< n \cdot f(A_{i_0^*}^0) + \frac{1 - \alpha_{i^*}^*}{2} \frac{f(S_{i^*}^*)}{2} + \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \\ &< n \cdot \frac{OPT}{4n \cdot m} + \frac{1 - \alpha_{i^*}^*}{2} \frac{f(S_{i^*}^*)}{2} + \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \\ &\leq \frac{(1 - \alpha_{i^*}^*) f(S_{i^*}^*)}{4} + \frac{1 - \alpha_{i^*}^*}{2} \frac{f(S_{i^*}^*)}{2} + \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*) \\ &= \alpha_{i^*}^* f(S_{i^*}^*) - c(S_{i^*}^*), \end{aligned}$$

where the first inequality is by subadditivity; the second inequality is since $\alpha_{i^*} \leq 1$; the fourth inequality is since agents that are not i^* don't take costly actions since their contract is 0 and by monotonicity of f ; the fifth

inequality is by subadditivity of f , and by our assumption towards contradiction; the sixth inequality holds since we consider the case where $f(A_{i_0^*}^0) < \frac{OPT}{4n \cdot m}$; the seventh inequality is by Proposition 7.1.

Thus, it holds that $f(S) \geq \frac{f(S_{i^*}^*)}{2}$, and the principal's utility is at least

$$(1 - \alpha_{i^*})f(S) \geq \frac{1 - \alpha_{i^*}}{2} \frac{f(S_{i^*}^*)}{2} = \Omega\left(\frac{OPT}{n \cdot m}\right),$$

which concludes the proof. $\square$

Finally, we show that for subadditive reward functions (or even XOS) the gap between the welfare and the worst case equilibrium under any contract can be quadratic.

EXAMPLE 7.2. Consider an instance with $n = 2k$ agents denoted by $[n]$, and binary actions. The reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ is

$$f(S) = \frac{\max\{|S|, k^2 \cdot |S \setminus [k]|, 2 \cdot \mathbf{1}[S \neq \emptyset]\}}{k^3}.$$

The function is XOS since it is the maximum over the additive functions $\frac{|S|}{k^3}$, $\frac{k^2 \cdot |S \setminus [k]|}{k^3}$, and the function $g_i(S) = \frac{2 \cdot \mathbf{1}[i \in S]}{k^3}$ for every i . The costs of exerting effort for agents in $[k]$ are 0, and the costs of agents in $[n] \setminus [k]$ are $\frac{k^2 - k/2}{k^3}$.

The social welfare of this instance is when all agents exert effort, leading to a social welfare of

$$OPT = 1 - k \cdot \frac{k^2 - k/2}{k^3} = \frac{1}{2k}.$$

On the other hand, let $\vec{\alpha}$ be some contract. If for all agents $i \in [n] \setminus [k]$ it holds that $\alpha_i < \frac{2k-1}{2k}$, then the set $\{1\}$ is an equilibrium. To see this, observe that agent $i \in A \setminus [k]$ does not want to deviate since

$$\alpha_i \cdot f(\{i, 1\}) - c(i) \leq \frac{2k-1}{2k} \frac{1}{k^2} - \frac{k^2 - k/2}{k^3} = 0 \leq \alpha_i \cdot f(1).$$

Agent $i \in [k] \setminus \{1\}$ does not want to deviate since

$$\alpha_i \cdot f(\{i, 1\}) - c(i) = \alpha_i \cdot f(1).$$

Thus, in this case, there is an equilibrium with a principal's utility of at most $f(1) = \frac{2}{k^3} = O\left(\frac{OPT}{n^2}\right)$.

If exists an agent $i \in [n] \setminus [k]$ with $\alpha_i \geq \frac{2k-1}{2k}$ then, there must be at most one such agent. Now consider the equilibrium where all agents in $[k]$ exert effort. Then agent i (and also all agents in $[n] \setminus [k]$) don't want to exert effort since

$$\alpha_i f([k]) = \alpha_i \cdot \frac{1}{2k^2} > \alpha_i \cdot \frac{1}{k} - \frac{k^2 - k/2}{k^3} = \alpha_i f([k] \cup \{i\}) - c(i),$$

where the inequality is since $\alpha_i \leq 1$. Now, the utility of the principal under this equilibrium is at most

$$(1 - \alpha_i)f([k]) \leq \frac{1}{2k} \cdot \frac{1}{k^2} = O\left(\frac{OPT}{n^2}\right).$$

8 Conclusion and Open Questions

In this work, we study a natural multi-agent combinatorial contracting problem with binary outcome. As our main result, we show that for submodular success probability f , it is possible to compute a contract whose worst equilibrium is a constant-factor approximation to the best equilibrium under any contract. A natural question is whether this result extends to XOS success probability functions.

We observe that our result, in its current form, cannot apply to XOS functions, even with binary actions. To see this, consider the instance with $n = 2k$ agents in Example 7.2. We already argued that in this example every contract $\vec{\alpha}$ admits an equilibrium with expected principal utility no more than $2/k^3$. On the other hand, consider contract $\vec{\alpha}$ such that $\alpha_1 = \dots = \alpha_{n-1} = 0$, $\alpha_n = 1 - \frac{1}{2k}$. One can verify that for this contract $\vec{\alpha}$ the set

$S = \{n\}$ is an equilibrium, and attains an expected principal utility of $1/(2k^2)$. Thus, for any contract the worst equilibrium is a factor $\Omega(n)$ worse than the best equilibrium of the best contract.

We could still hope for a constant-factor approximation for XOS, in the following two weaker senses. We could either aim for a contract together with a recommended equilibrium for that contract that approximates the best equilibrium under any contract (as done for binary actions and XOS rewards in [19]), or we could change the benchmark to be the best contract measured in terms of worst-case equilibrium.

Our current approach uses submodularity in various places. In particular, the Doubling Lemma heavily relies on the function f being submodular. Similarly, the sufficient condition (Lemma 3.2) uses submodularity, as do the lemmas for finding a set that satisfies this condition (Lemmas 3.3 and 3.4). In contrast, Lemma 3.6 does apply also to XOS functions, and we used submodularity just for simplicity (since the proof for XOS is more complex).

Another interesting question is whether the multi-agent combinatorial contracting problem admits a PTAS/FPTAS when the rewards are gross substitutes. This problem is open even for the binary-action case. Our work implies a constant-factor approximation with value oracle access, but our hardness result that rules out a PTAS is for submodular rewards.

It would also be interesting to explore the multi-agent combinatorial contracting problems with a constant number of actions per agent or with a constant number of agents. For example, when each agent has a constant number of actions, is it possible to get a constant-factor approximation for submodular success probabilities without demand queries? As we have shown, we cannot hope to get a better than constant-factor approximation, even in the case of binary actions, with access to both value and demand queries. On the other hand, as was shown in [19], it is possible to obtain a constant-factor approximation in the binary-action case with value queries only. With a general number of actions, we cannot hope to get a constant-factor approximation with value queries only, even for a single agent (assuming $P \neq NP$) [27].

Interesting questions for a setting with a constant (or logarithmic) number of agents include: Can we get the exact optimum for gross substitutes rewards (as done in [18] for a single agent), a PTAS/FPTAS for submodular rewards, or a constant-factor approximation for subadditive rewards (all with value and demand oracle access)?

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A Proof of Proposition 2.1

Fix $\vec{\alpha} \in [0, 1]^n$. Consider $\phi(S) : 2^A \rightarrow \mathbb{R}$, with $\phi(S) = f(S) - \sum_{i \in [n]} \sum_{j \in S_i} \frac{c_j}{\alpha_i}$. If $\alpha_i = 0$, we define $c_j/\alpha_i = \infty$ when $c_j > 0$, and we let $c_j/\alpha_i = 0$ when $c_j = 0$.

Note that when $\alpha_i = 0$, any set of actions $S_i \subseteq A_i$ that contains an item $j \in S_j$ such that $c_j > 0$ is strictly dominated (e.g., by the empty set $\emptyset$). We thus assume that for agents i such that $\alpha_i = 0$ all actions in A_i have cost $c_j = 0$. Note that this in particular means that agent i receives a utility of zero for all possible sets of actions $S_i \subseteq A_i$.

This is a somewhat subtle point, because although the agent with $\alpha_i = 0$ is indifferent which set of actions with zero cost he chooses, the principal derives different rewards from different choices; and also the strategic choices of the other agents are influenced (through the payments they receive which depend on the principal's reward).

CLAIM A.1. *Function $\phi : 2^A \rightarrow \mathbb{R}$ is a generalized ordinal potential function.*

Proof. Recall that agent i 's utility is $u_i(S, \vec{\alpha}) = \alpha_i f(S) - \sum_{j \in S_i} c_j$. We need to show that for any i, S_i, S'_i, S_{-i}

$$u_i(S_i, S_{-i}, \vec{\alpha}) > u_i(S'_i, S_{-i}, \vec{\alpha}) \implies \phi(S_i, S_{-i}) > \phi(S'_i, S_{-i}).$$

We only need to consider i such that $\alpha_i > 0$. Note that, since $\alpha_i > 0$, if $u_i(S_i, S_{-i}, \vec{\alpha}) > u_i(S'_i, S_{-i}, \vec{\alpha})$, then

$$f(S_i, S_{-i}) - \sum_{j \in S_i} \frac{c_j}{\alpha_i} > f(S'_i, S_{-i}) - \sum_{j \in S'_i} \frac{c_j}{\alpha_i}.$$

Thus, $u_i(S_i, S_{-i}, \vec{\alpha}) > u_i(S'_i, S_{-i}, \vec{\alpha})$ implies that

$$\phi(S_i, S_{-i}) - \phi(S'_i, S_{-i}) = \left(f(S_i, S_{-i}) - \sum_{j \in S_i} \frac{c_j}{\alpha_i} \right) - \left(f(S'_i, S_{-i}) - \sum_{j \in S'_i} \frac{c_j}{\alpha_i} \right) > 0,$$

as claimed. $\square$

Proposition 2.1 follows from Lemma 2.5 of [41] which shows that any finite game that admits a generalized ordinal potential function has the finite improvement property, and hence at least one (pure) Nash equilibrium.

B Demand Queries and Equilibria

Fix $\vec{\alpha} \in [0, 1]^n$. Recall the definition of the potential function $\phi(S) : 2^A \rightarrow \mathbb{R}$, with $\phi(S) = f(S) - \sum_{i \in [n]} \sum_{j \in S_i} \frac{c_j}{\alpha_i}$. Consider a demand query with respect to f at prices $p_j = c_j/\alpha_i$ for $j \in A_i$ such that $\alpha_i > 0$, and $p_j = 0$ for $j \in A_i$ such that $c_i = 0$.

As discussed in Appendix A, We can assume that for agents such that $\alpha_i = 0$, the action sets A_i are pruned so that $c_j = 0$ for all $j \in A_i$.

CLAIM B.1. *Any $S \subseteq A$ that maximizes $\phi(S)$ is a (pure) Nash equilibrium.*

Proof. Note that if $S \subseteq A$ maximizes $\phi(S)$, then since ϕ is a generalized ordinal potential function (Claim A.1), if for some agent i there existed a S'_i such that $u_i(S'_i, S_{-i}, \vec{\alpha}) > u_i(S_i, S_{-i}, \vec{\alpha})$, then we would also have $\phi(S'_i, S_{-i}) > \phi(S_i, S_{-i})$, in contradiction to S being a maximum of ϕ . So, for all i and all S'_i , we must have $u_i(S_i, S_{-i}, \vec{\alpha}) \geq u_i(S'_i, S_{-i}, \vec{\alpha})$.

Note that, in particular, since for every i and any S'_{-i} , it holds that $u_i(\emptyset, S'_{-i}, \vec{\alpha}) = \alpha_i f(\emptyset, S'_{-i}) \geq 0$, we must have $u_i(S_i, S'_{-i}) \geq 0$. $\square$

More generally, there is a one-to-one correspondence between equilibria and local optima of the potential function, where “local” refers to any change in actions controlled by a single agent.

We next show that when the reward function is XOS, then there is an instance in which for every contract, there is a bad equilibrium. This is true even for the binary-action case.

PROPOSITION B.1. *There exists an instance with XOS reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ and costs c , for which for every contract $\vec{\alpha}$ there exists an equilibrium S such that the principal's utility is at most $O\left(\frac{u_p(S^*, \vec{\alpha}^*)}{|A|}\right)$, where $\vec{\alpha}^*, S^*$ are the best contract with the best corresponding equilibrium.*

Proof. Consider an instance with $n > 2$ agents with binary actions where the reward function $f : 2^A \rightarrow \mathbb{R}_{\geq 0}$ is

$$f(S) = \begin{cases} \frac{2}{n} & \text{if } |S| = 1 \\ \frac{|S|}{n} & \text{otherwise} \end{cases}$$

and let the cost of all agents be $c_i = \frac{1}{2 \cdot n^2}$. The reward function is XOS since it is the maximum over the additive function $g(S) = \frac{|S|}{n}$ and the functions $g_i(S) = \frac{2 \cdot \mathbf{1}[i \in S]}{n}$ for every agent i . The optimal contract $\vec{\alpha}^*$ is to pay each agent $\alpha_i = \frac{1}{2n}$ of the reward for which the optimal equilibrium is $S^* = A$, which leads to a principal utility of $u_p(S^*, \vec{\alpha}^*) = (1 - n \cdot \frac{1}{2n}) \cdot \frac{n+0}{n} = \frac{1}{2}$. However, for every contract $\vec{\alpha}$ there exists an equilibrium in which at most one agent exerts effort.

To see this, let S be a set of agents of size at least two which is an equilibrium with respect to $\vec{\alpha}$. (We know that an equilibrium exists, and if all equilibria are at most of size one, then we are done.) It means that for every $i \in S$ it holds that $\alpha_i f(i \mid S \setminus \{i\}) \geq c_i$ as otherwise the agent would prefer to not exert effort. By rearranging we get that $\alpha_i \geq \frac{c_i}{f(i \mid S \setminus \{i\})} \geq \frac{1/2n^2}{1/n} = \frac{1}{2n}$. Let i^* be an arbitrary agent in S (such exists since $|S| \geq 2$). We claim that the set $S' = \{i^*\}$ is also an equilibrium with respect to $\vec{\alpha}$. This is true since for agent i^* it holds that $\alpha_{i^*} f(i^* \mid S' \setminus \{i^*\}) = \alpha_{i^*} \frac{2}{n} \geq \frac{1}{2n^2} = c_{i^*}$. For every agent $i \neq i^*$ it holds that $f(\{i, i^*\}) = \frac{2}{n} = f(\{i^*\})$ and so it holds that $\alpha_i f(i \mid S' \setminus \{i\}) = \alpha_{i^*} 0 \leq \frac{1}{2n^2} = c_{i^*}$.

Thus, any contract has an equilibrium that incentivizes at most one agent, which leads to a utility of at most $\frac{2}{n}$. $\square$

PROPOSITION B.2. *For every instance defined by $f : 2^A \rightarrow \mathbb{R}_{\geq 0}, c$, the supremum is attained at some contract $\vec{\alpha}^*$ with a corresponding set S^* .*

Proof. For every set of actions S , the set of contracts that incentivizes it $T_S = \{\vec{\alpha} \mid S \in \text{NE}(\vec{\alpha})\}$ is the set of actions that satisfies for every i and $S'_i \subseteq A_i$ that $\alpha_i f(S) - c(S_i) \geq \alpha_i f(S'_i, S_{-i}) - c(S'_i)$. Since when fixing the actions of the agents, the principal's utility is a continuous function, and the set T_S is compact, we get by the Extreme Value Theorem that there exists a contract $\vec{\alpha}_S \in T_S$ for which

$$u_p(S, \vec{\alpha}_S) = \sup_{\vec{\alpha}} u_p(S, \vec{\alpha}).$$

Let $S^* = \arg \max u_p(S, \vec{\alpha}_S)$. By definition, $S^* \in \text{NE}(\vec{\alpha}_{S^*})$. Then, for every α , and $S \in \text{NE}(\vec{\alpha})$ it holds that

$$u_p(S, \vec{\alpha}) \leq u_p(S, \vec{\alpha}_S) \leq u_p(S^*, \vec{\alpha}_{S^*}),$$

which concludes the proof. $\square$